Broadcast Technology UG1

4: Electronics for Broadcast – Review

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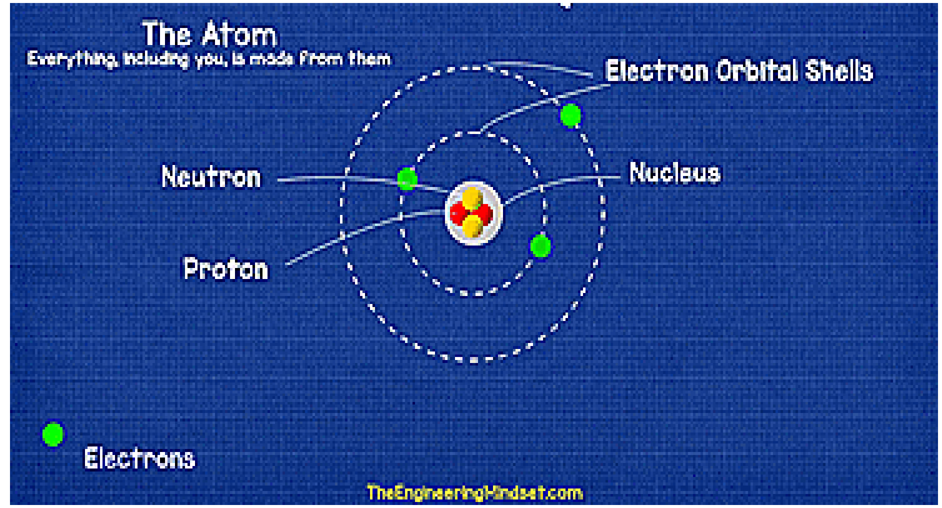
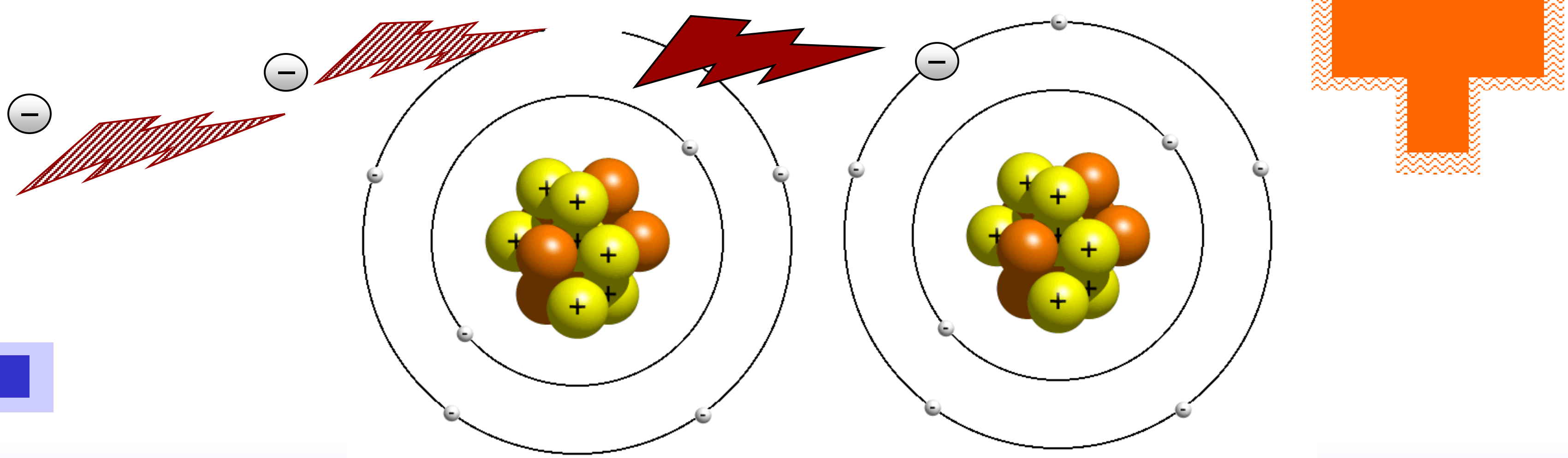
shielding, ground
coax
impedance
characteristic impedance

- Earth, Chassis, etc
- Circuit Interconnects, Interference
- Shielding
- Grounding/Earthing
- Unbalanced Connections
- Ground Loops
- Twisted Pair Cable
- Balanced Lines
- Balanced Connectors
- Unbalanced-To- Balanced
- Balanced -To- Unbalanced
- Phantom Power
- Line Level
- Multistages
- Impedance
- Characteristic Impedance

Conduction

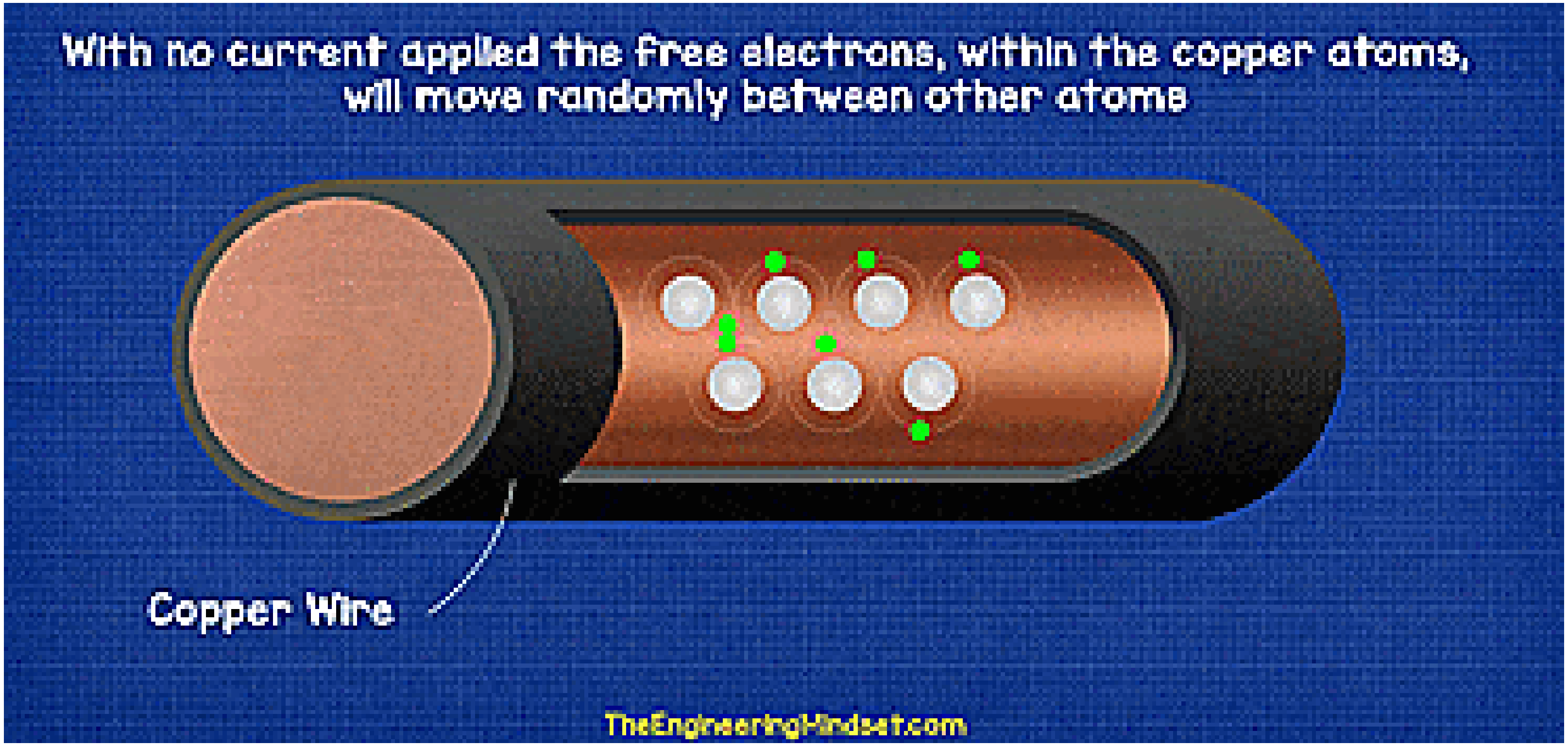
- At the centre of an Atom is the Nucleus, this houses two particles known as the Neutron and Proton.
- Neutron has no electrical charge but the Proton has a positive electrical charge.
- Circling the nucleus is a cloud of Electrons, which are negatively charged.
- Conductors (metals, e.g. Copper) – have loose bindings to their outer electrons





Conduction

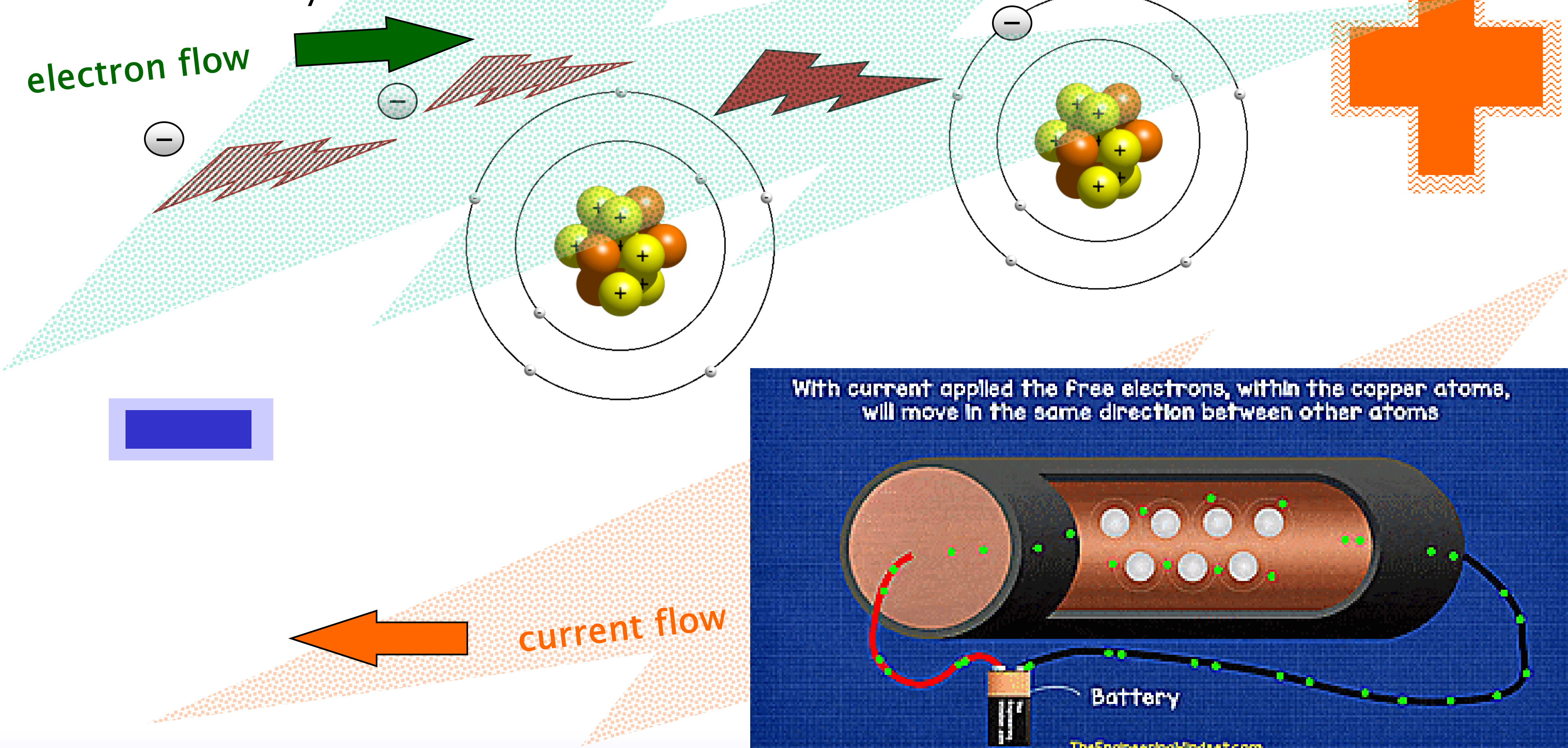
No field – Free electrons of the atoms move from one atom to another. This occurs randomly in any direction.
... net current flow is zero



Conduction

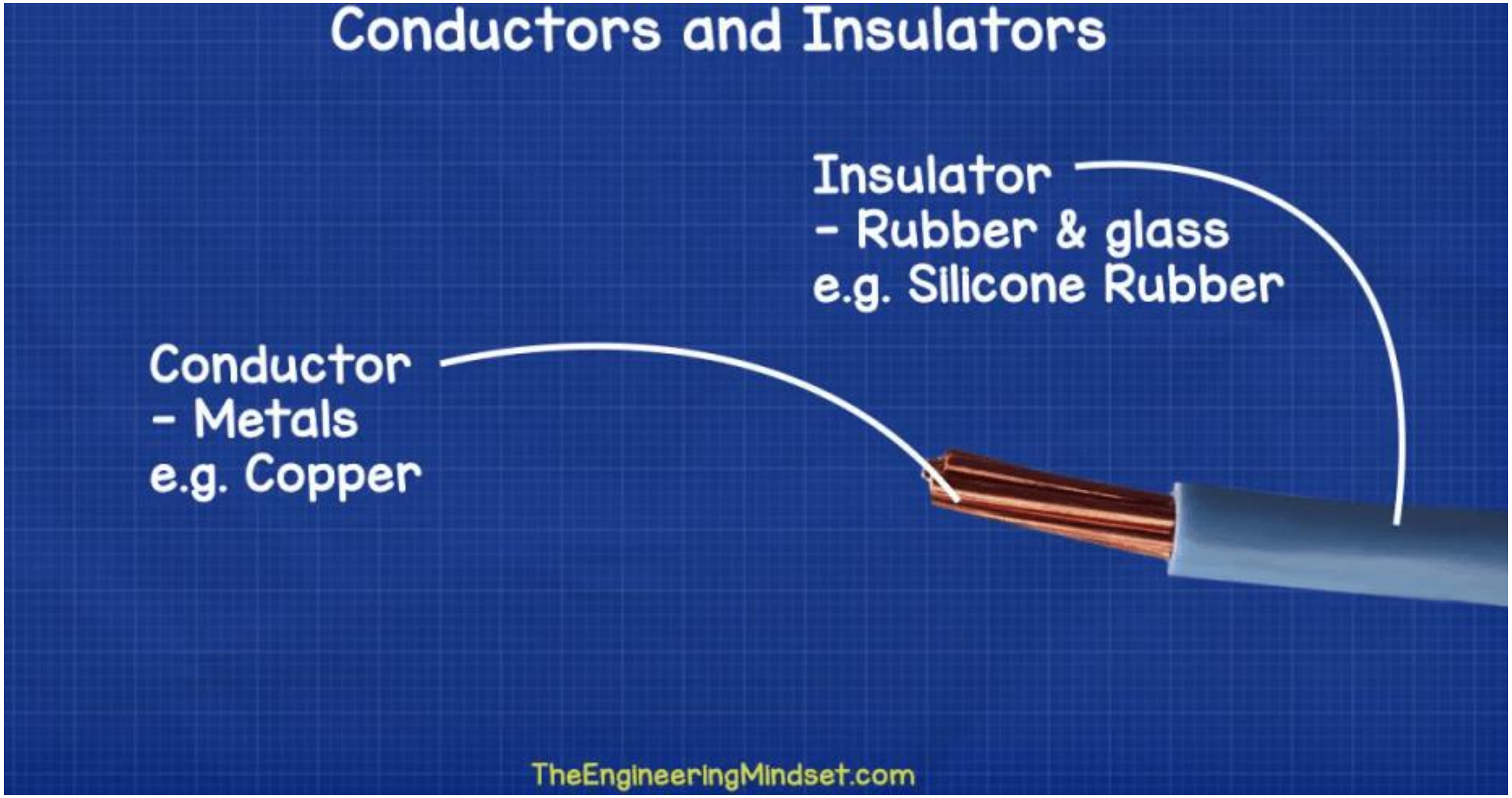
- When an electric field is applied, these free electrons feel a force.
- They are attracted toward the positive electrode, drifting through the material in that direction.
- Movement of electrons in ONE direction gives rise to ELECTRIC current

  ... is really a movement of CHARGE



Insulators

- Atoms which do not have free electrons are known as insulators
- restrict movement of electrons
- electrons firmly bound to atom



Shure SM58 Specifications

http://www.shure.com/idc/groups/public/documents/webcontent/us_pro_sm58_specsheet.pdf

Product Specifications

SM58® Cardioid Dynamic Microphone

SM58

Overview

The legendary SM58® is an industry-standard, highly versatile cardioid dynamic vocal microphone that is consistently the first choice of vocal performers around the globe. Even in extreme conditions, the SM58 is tailored to target the main sound source while minimizing background noise, delivering warm and clear vocal reproduction.

Features

- Frequency response tailored for vocals, with brightened midrange and bass rolloff
- Uniform cardioid pickup pattern isolates the main sound source and minimizes background noise
- Pneumatic shock-mount system cuts down handling noise
- Effective, built-in spherical wind and pop filter
- Supplied with break-resistant stand adapter which rotates 180 degrees
- Legendary Shure quality, ruggedness and reliability
- Cardioid (unidirectional) dynamic
- Frequency response: 50 to 15,000 Hz

Available Models

SM58-LC	Includes Stand Adapter and Zippered Pouch
SM58-CN	Includes 7.6 m (25 ft) XLR-Male to XLR-Female Cable, Swivel Adapter and a Zippered Pouch
SM58S	Includes Integrated On/Off Switch, Swivel Adapter and a Zippered Pouch

Specifications

Type	Dynamic
Frequency Response	50 to 15,000 Hz
Polar Pattern	Cardioid
Sensitivity (at 1,000 Hz Open Circuit Voltage)	–54.5 dBV/Pa (1.85 mV) 1 Pa = 94 dB SPL
Impedance	Rated impedance is 150Ω (300Ω actual) for connection to microphone inputs rated low impedance
Polarity	Positive pressure on diaphragm produces positive voltage on pin 2 with respect to pin 3.
Case	Dark gray, enamel-painted, die cast metal; matte-finished, silver colored, spherical steel mesh grille
Connector	Three-pin professional audio connector (male XLR type)
Connector	Three-pin professional audio connector (male XLR type)
Net Weight	298 grams (10.5 oz)
Dimensions	162 mm (6-3/8 in.) L x 51 mm (2 in.) W

Output voltage ~ 0.19mV $= \frac{19}{1000}$ V

Shure SM58 Specifications

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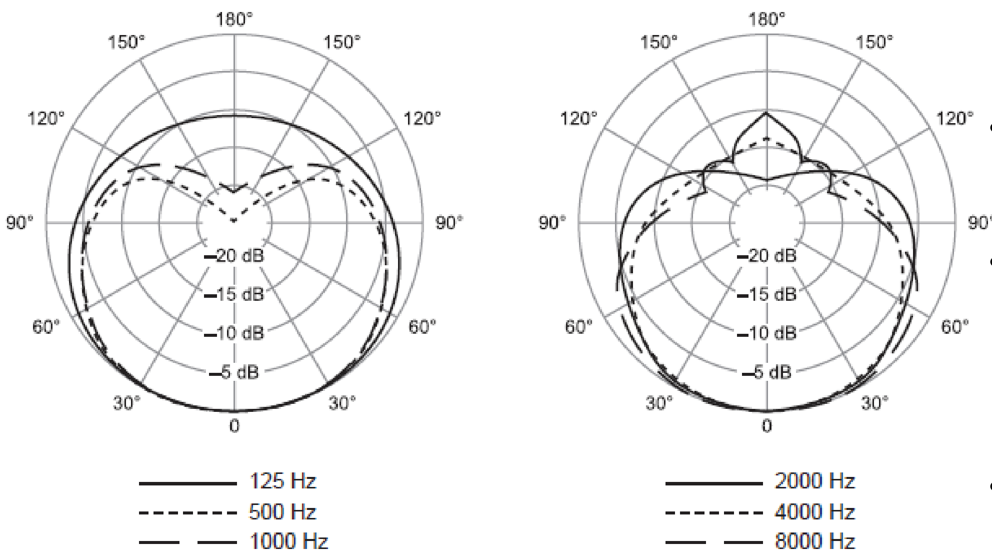
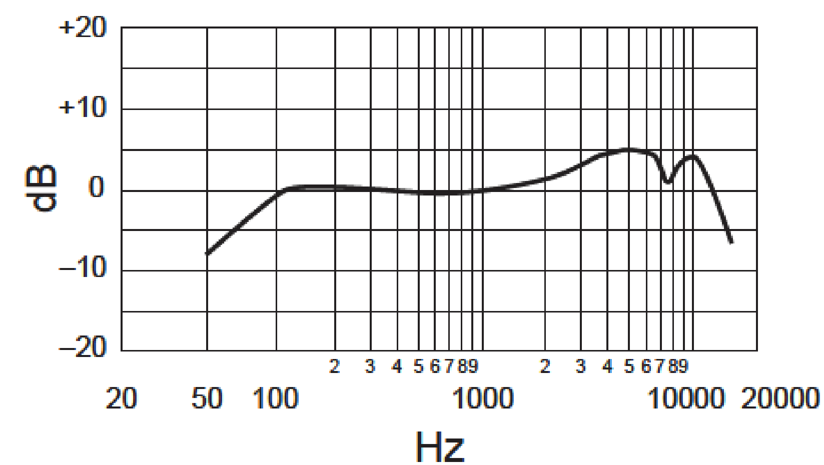


Polar Pattern

- The polar pattern illustrates how the microphone's sensitivity changes with the angle of the sound source.
- Angles marked from 0° (front of the mic) to 180° (rear), with 90° and 270° representing the sides.
- The deep null at 180° means excellent rejection of sound from the rear, making it more effective at isolating the sound source (e.g., a singer's voice) from background noise.
- As frequency increases (2 kHz to 8 kHz), the pattern narrows at the front and becomes more directional, with deeper nulls at the sides.

- The frequency response graph offers valuable insights into how the microphone performs across the spectrum (50 Hz to 15 kHz).

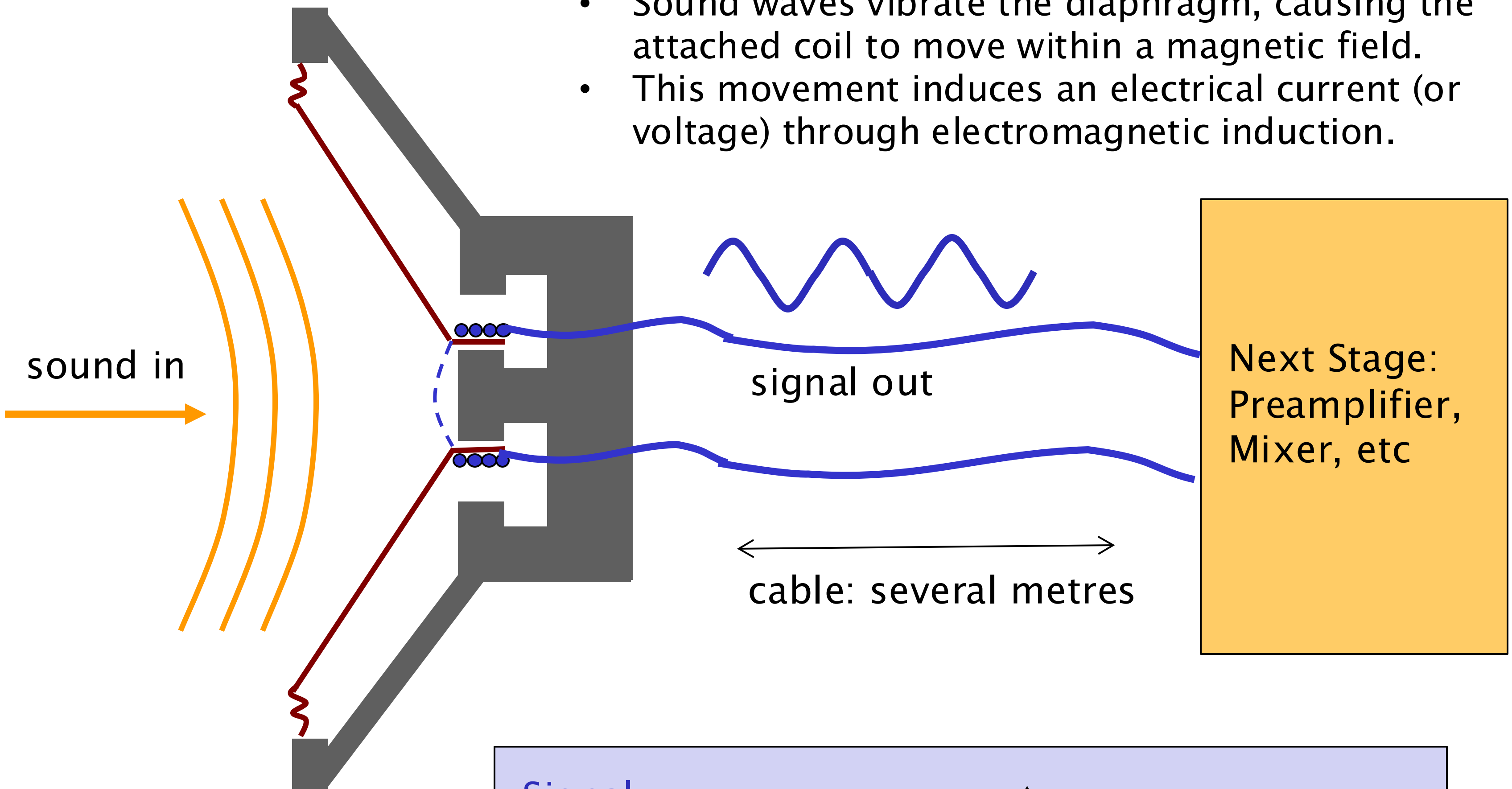
- As a vocal mic, we expect the SM58's frequency response to have smooth transitions without drastic peaks or dips that could make the sound unnatural.

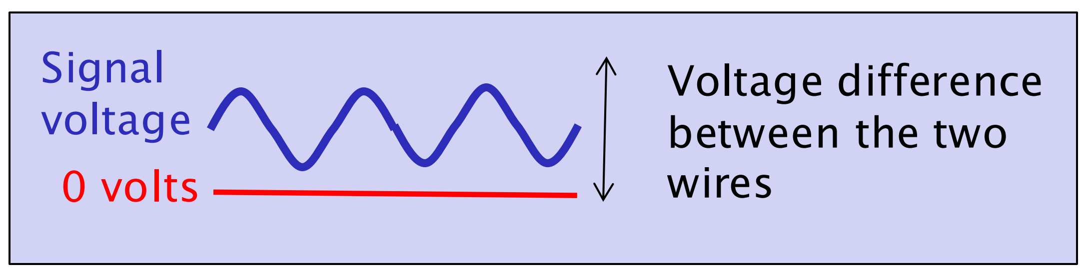


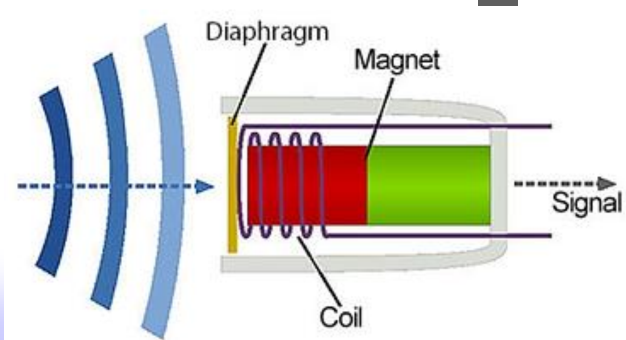
Frequency Response

Dynamic Microphones

- Dynamic microphones convert sound waves into an electrical signal, which is subsequently amplified.
- Sound waves vibrate the diaphragm, causing the attached coil to move within a magnetic field.
- This movement induces an electrical current (or voltage) through electromagnetic induction.

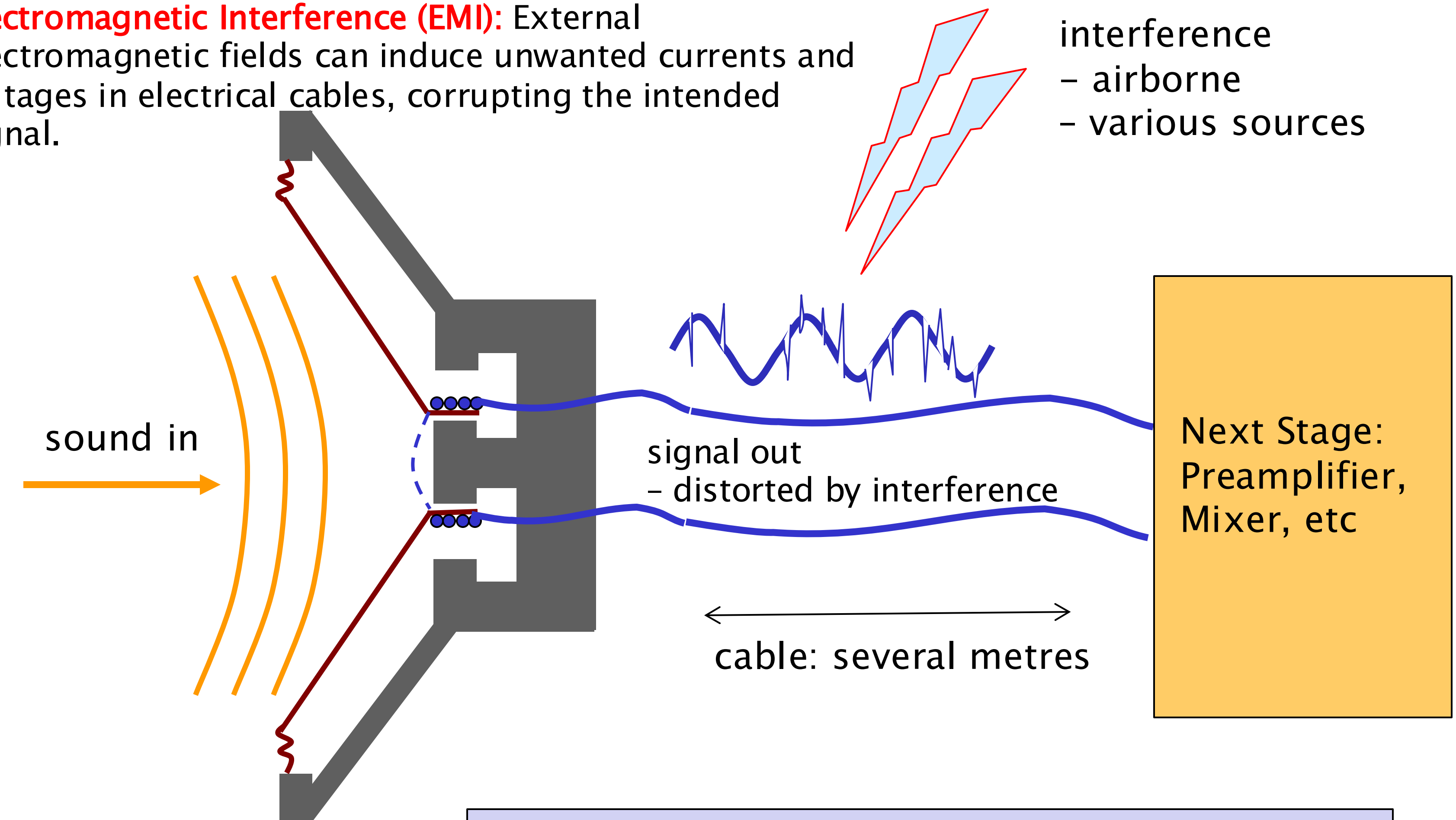




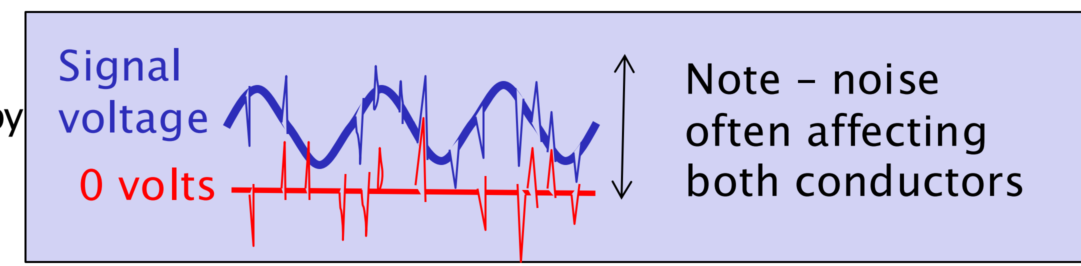


Interference Affects the Signal

- **Electromagnetic Interference (EMI):** External electromagnetic fields can induce unwanted currents and voltages in electrical cables, corrupting the intended signal.

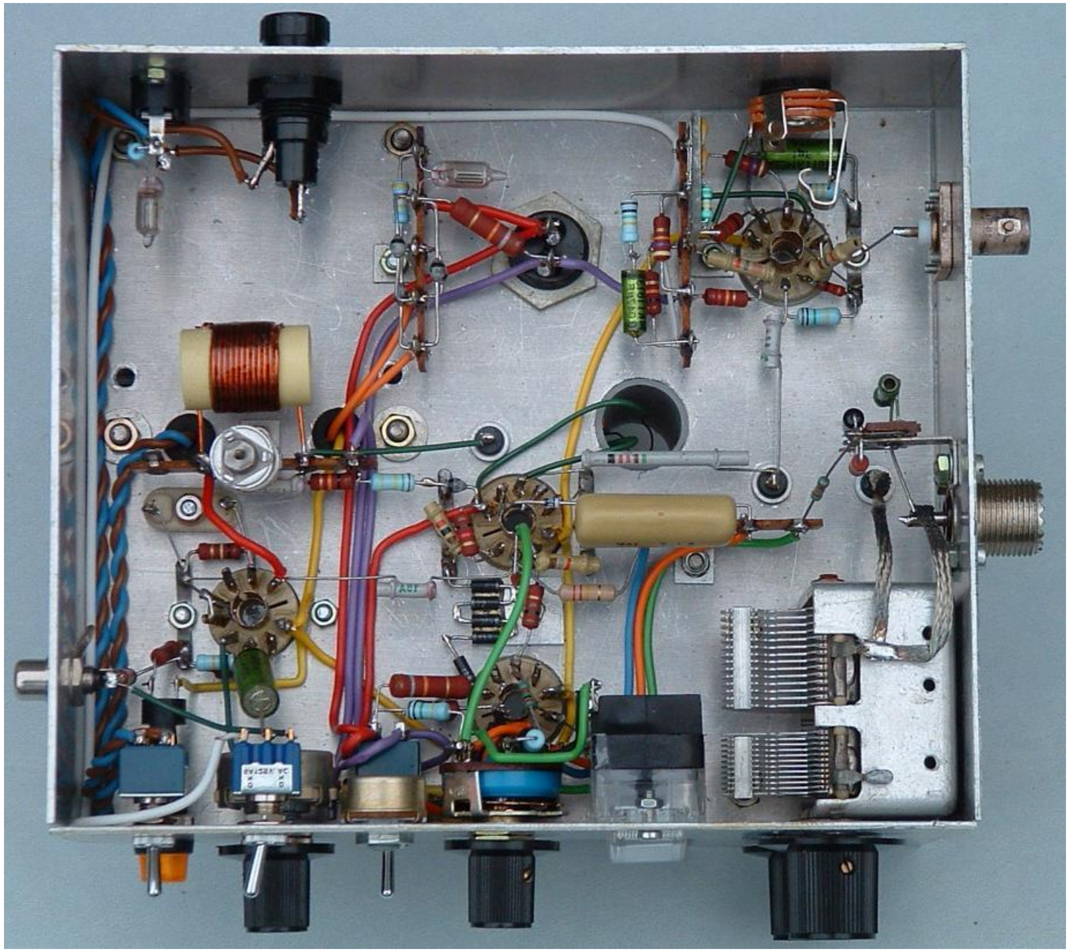


- As the "signal out" travels along the cable, it becomes corrupted by the interference.
- Interference degrades the quality of the audio signal, introducing unwanted noise and distortion.



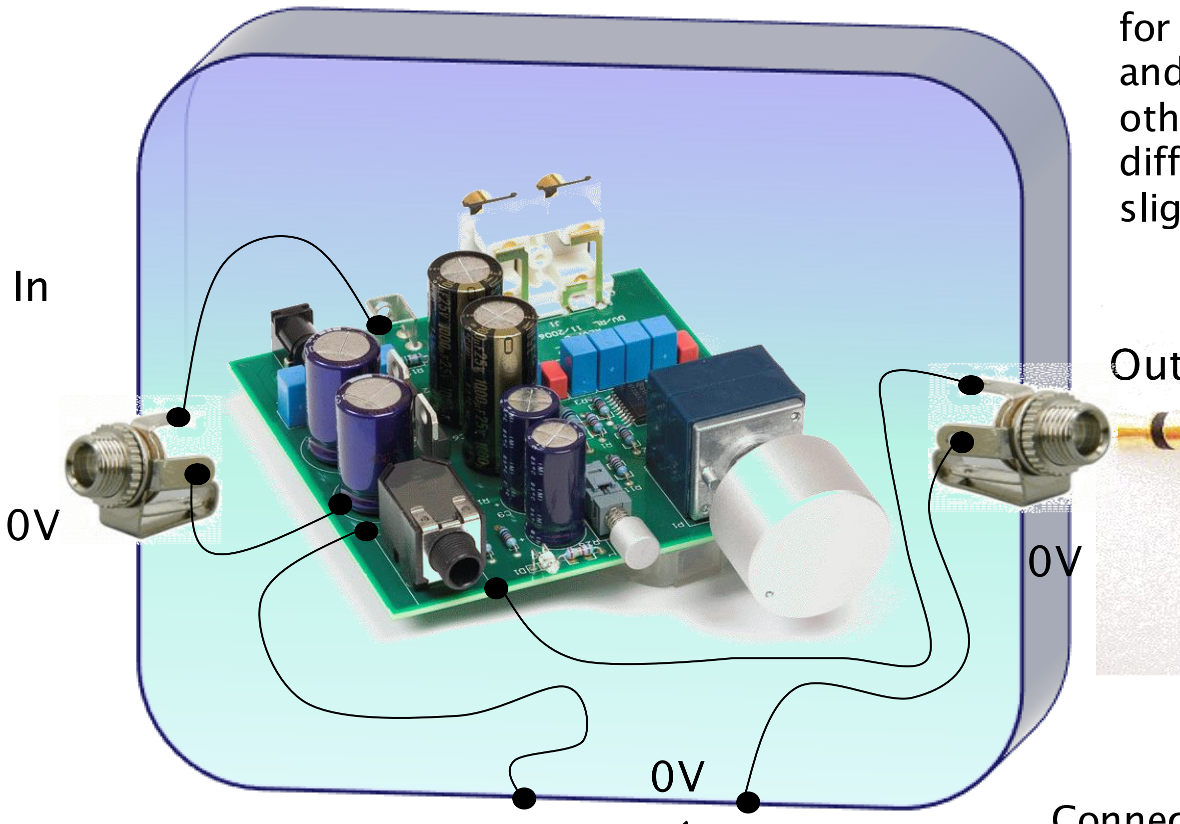
Chassis

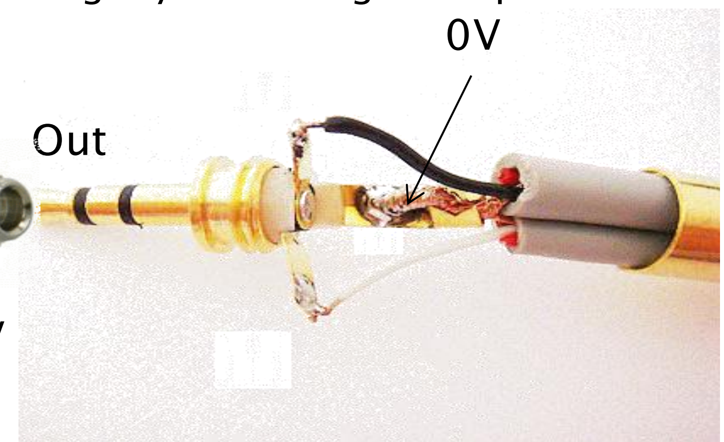
- box to contain and surround circuit
- electrically conductive- typically made of metal (like aluminium or steel)
- protects from external interference- acts as a Faraday cage
- needs chassis to be connected to a common zero volts level, or even earth



Chassis

Common Reference Potential:
Connecting all the 0V points to the chassis ensures that all parts of the circuit share the same ground reference potential. This is required for the correct operation of the circuit and helps to prevent ground loops and other noise issues that can arise from different parts of the circuit having slightly different ground potentials.

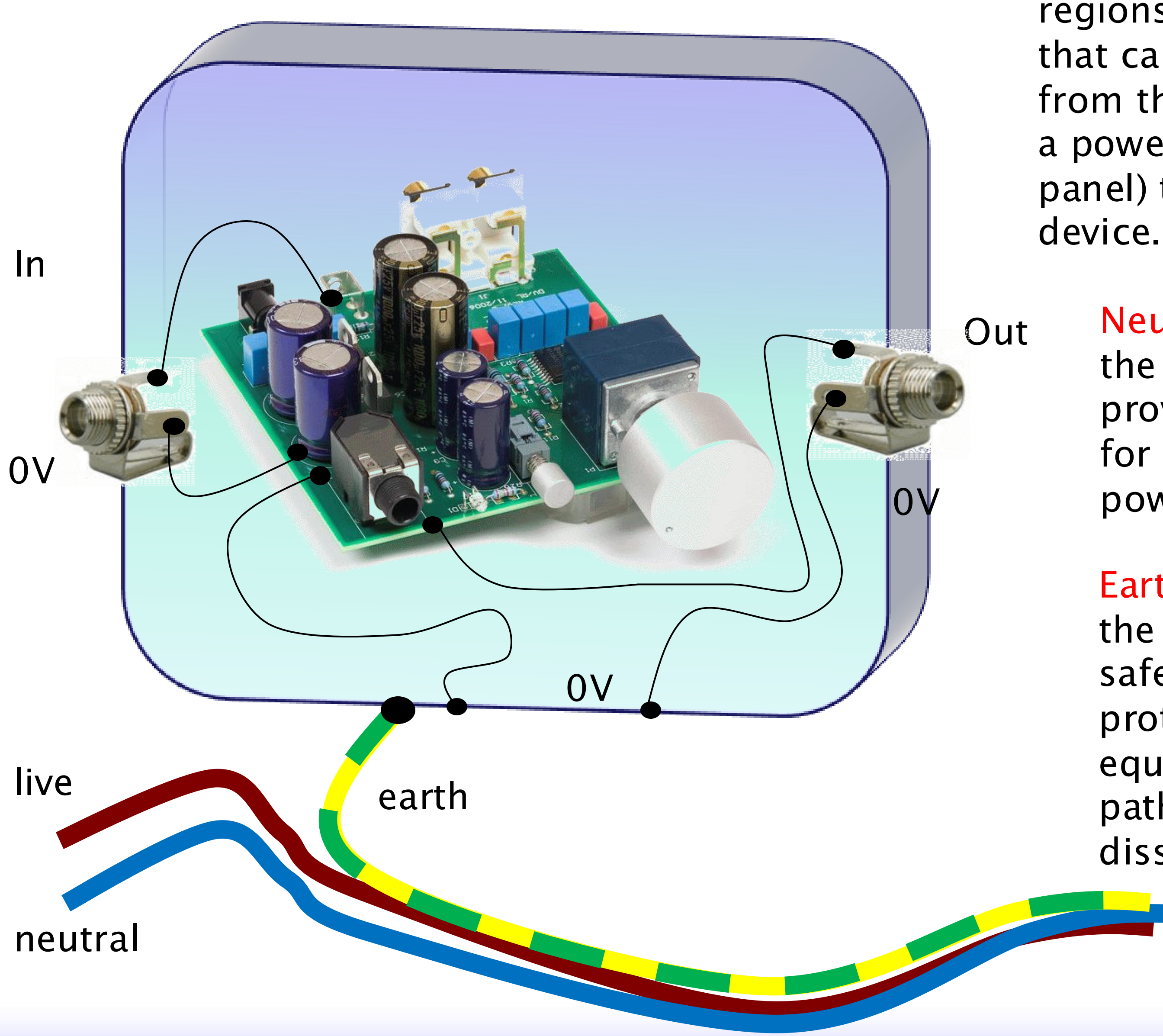




Connecting the chassis to ground (0V) helps shield the circuit from external electromagnetic interference (EMI).

Chassis – is connected to all 0V points in the circuit as a “common rail”

Earth



Live cable (sometimes called the "hot" wire in certain regions) is the conductor that carries electrical current from the power source (e.g., a power station or breaker panel) to the appliance or device.

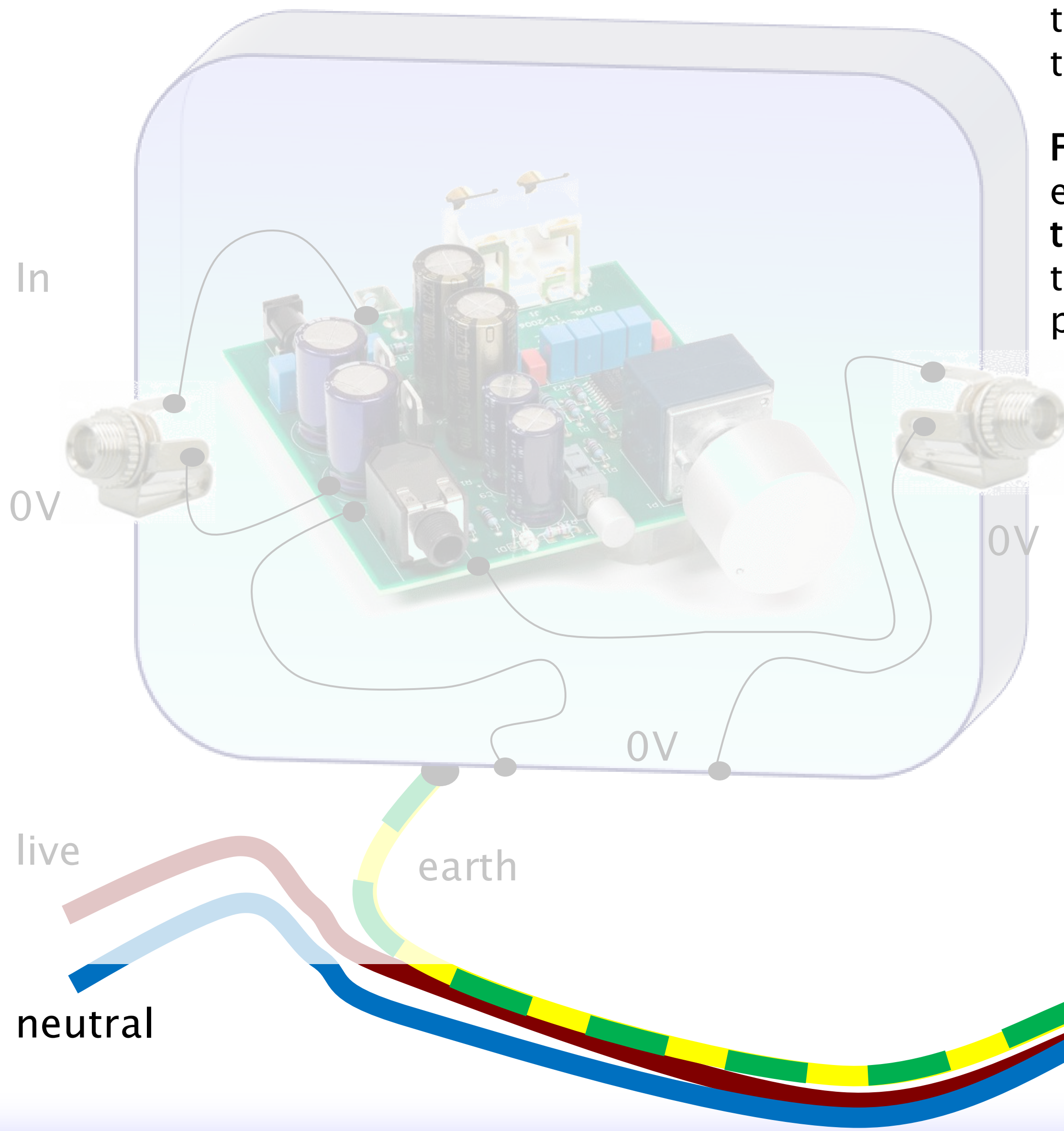
Neutral cable completes the electrical circuit by providing a return path for the current back to the power source.

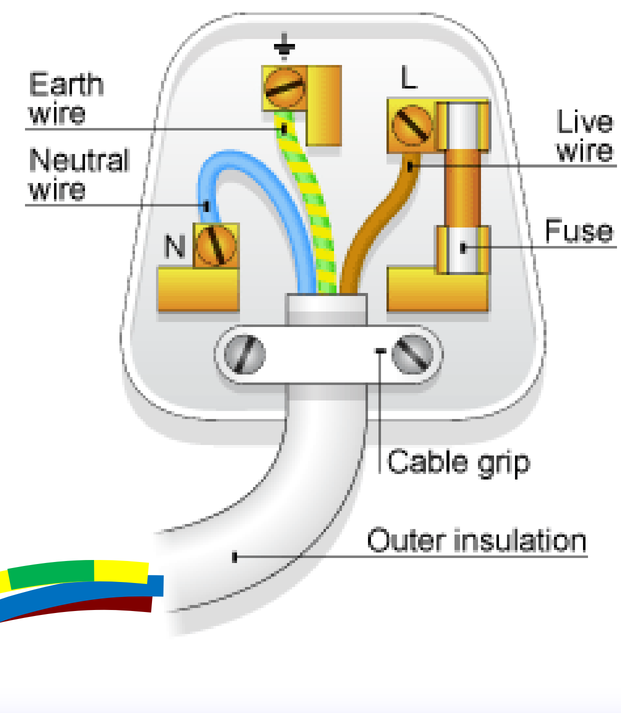
Earth cable (also called the ground wire) is a safety feature designed to protect users and equipment by providing a path for fault currents to dissipate into the ground.

Mains Plug Wiring

Fuse is connected in series with the **Live cable (Brown).** This means that all the electrical current flowing to the device through the Live cable must also pass through the fuse.

Fuse is a deliberate weak point in the electrical circuit. It is designed to **break the circuit** if the current flowing through it exceeds a certain predetermined level for too long.





Earth, Chassis

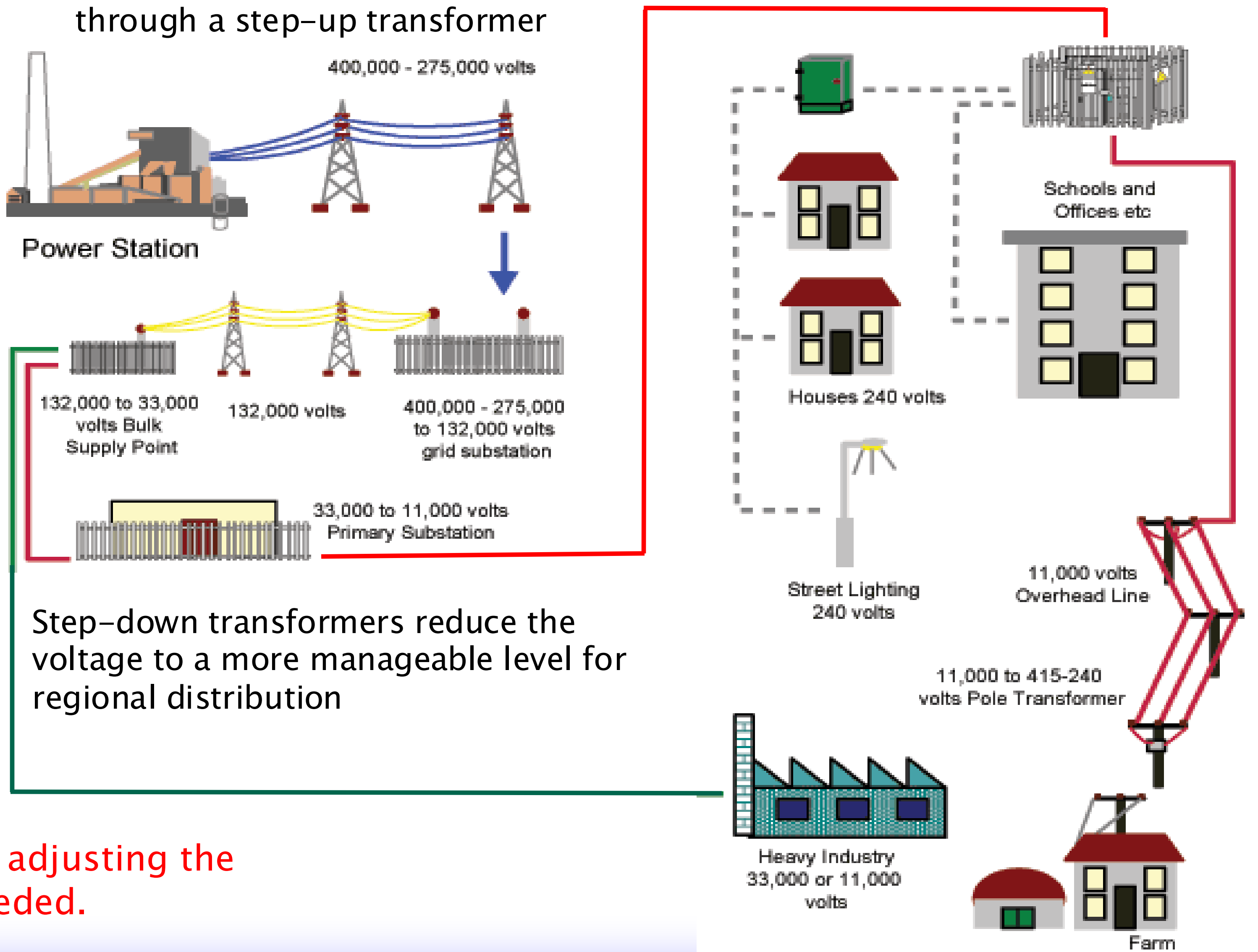
Earth connection

Power Distribution, using AC

Journey of electricity from the power station to the end users

Power stations produce electricity at a relatively low voltage, typically between 11 kV and 33 kV. This voltage is suitable for generation but too low for efficient long-distance transmission.

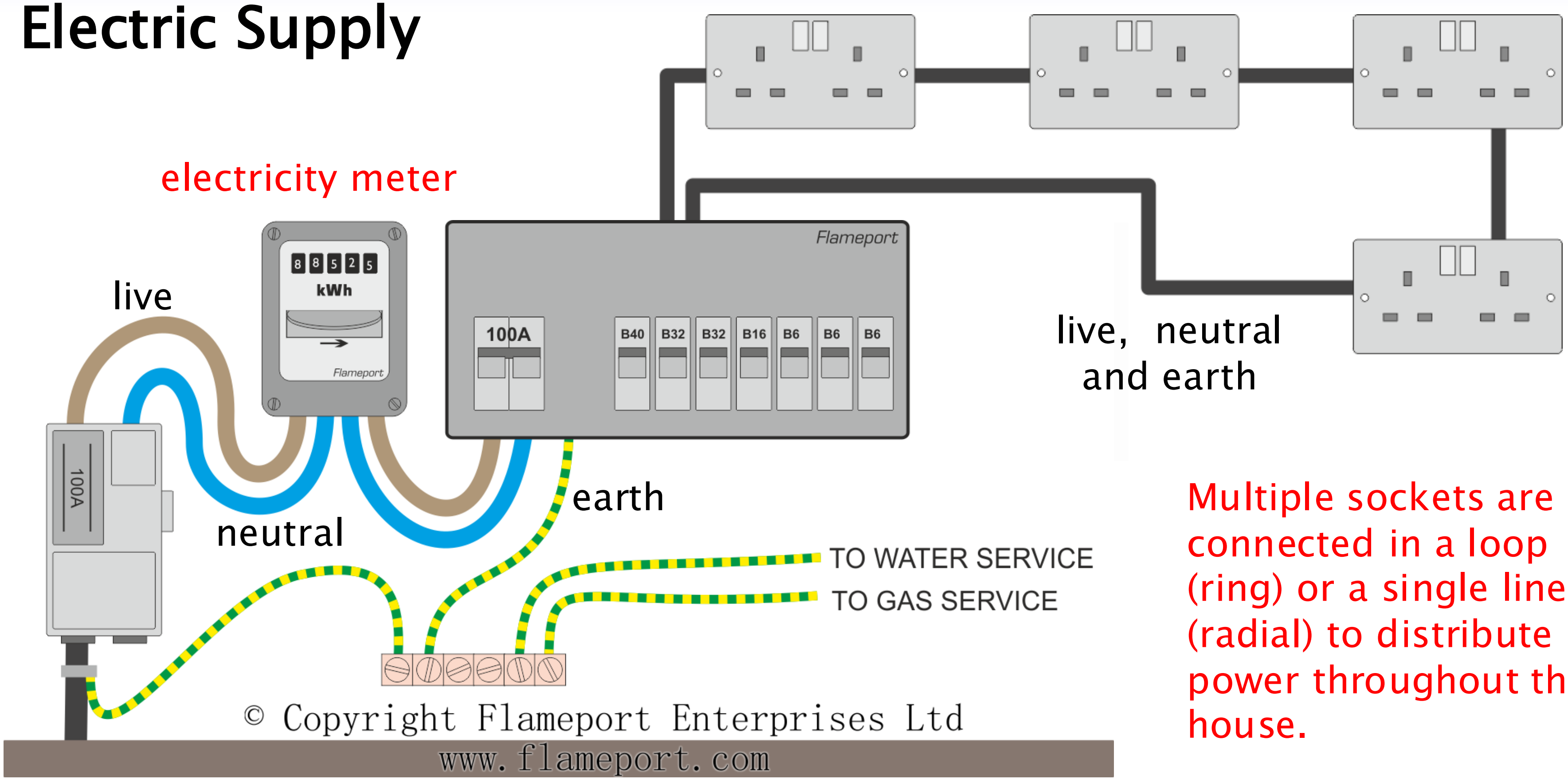
Before electricity is transmitted over long distances, it passes through a step-up transformer



Step-down transformers reduce the voltage to a more manageable level for regional distribution

Transformers adjusting the voltage as needed.

Electric Supply



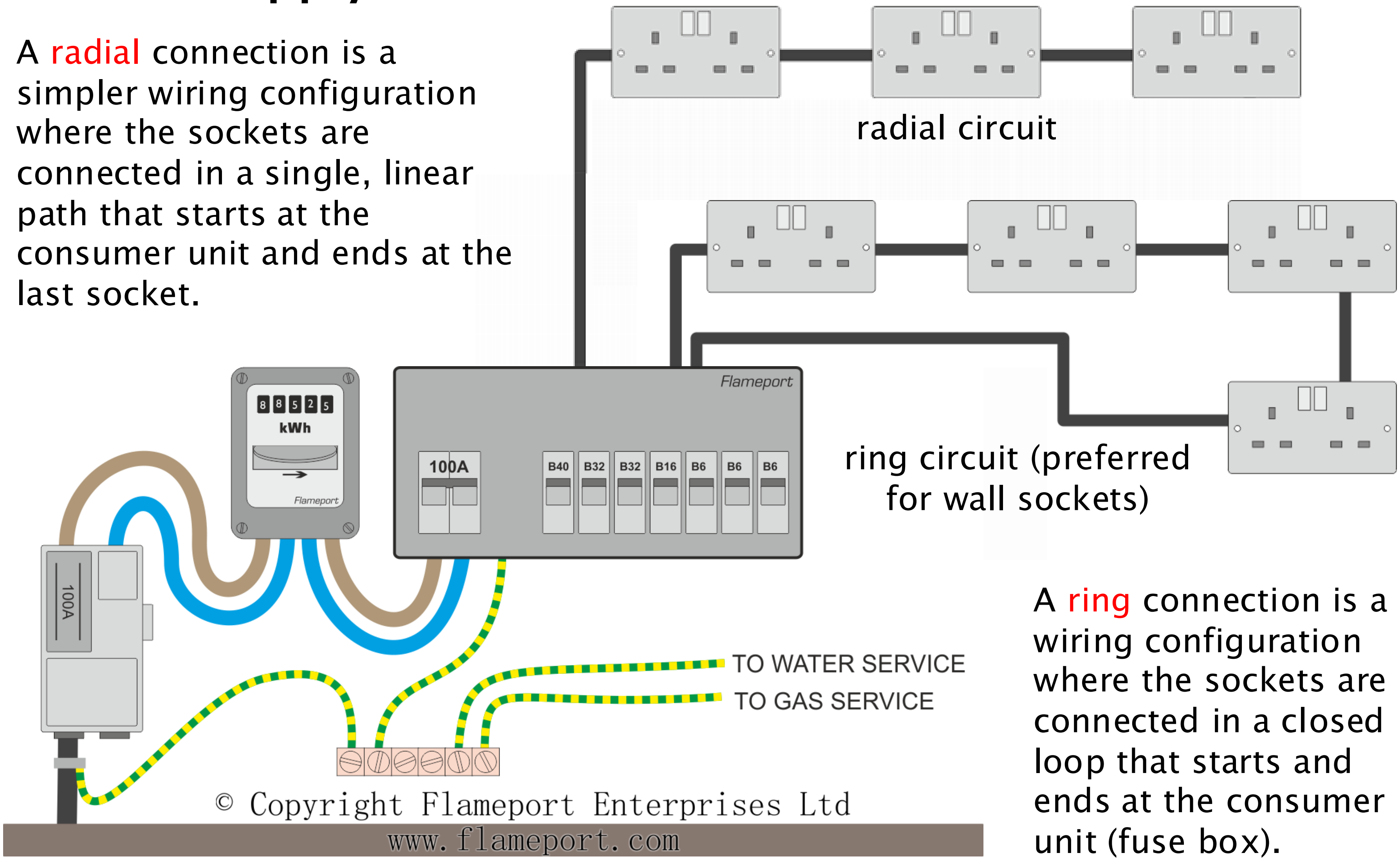
Multiple sockets are connected in a loop (ring) or a single line (radial) to distribute power throughout the house.

Meter receives 230V supply from electrical distribution network – live and neutral

- Consumer Unit (Fuse Board) is the central distribution point for electricity within the premises. It contains circuit breakers (or fuses) to protect individual circuits.
    - fused at different current ratings for ring main (pictured), lighting, heating, etc
- earth connected locally

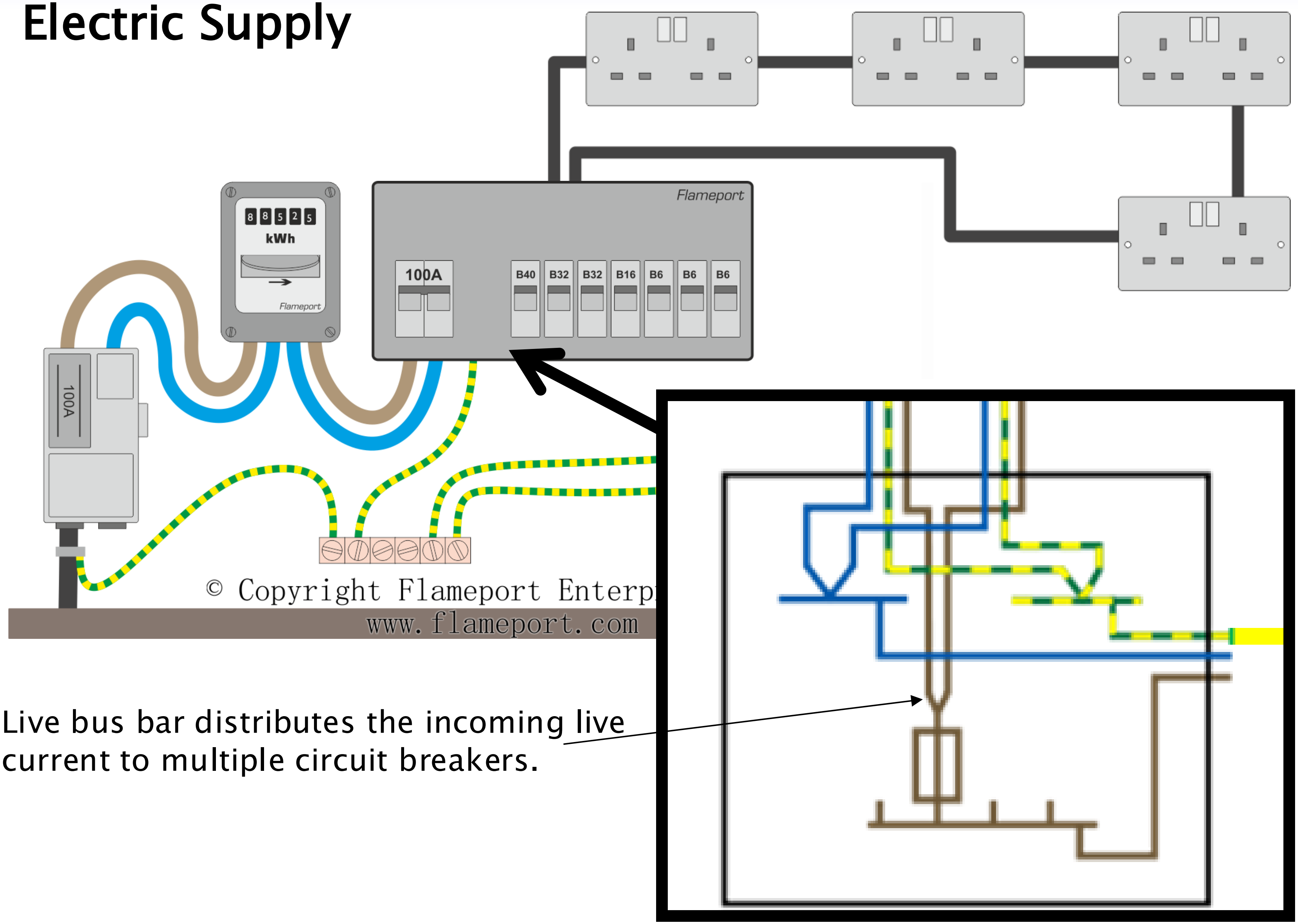
Electric Supply

A radial connection is a simpler wiring configuration where the sockets are connected in a single, linear path that starts at the consumer unit and ends at the last socket.



A ring connection is a wiring configuration where the sockets are connected in a closed loop that starts and ends at the consumer unit (fuse box).

Electric Supply

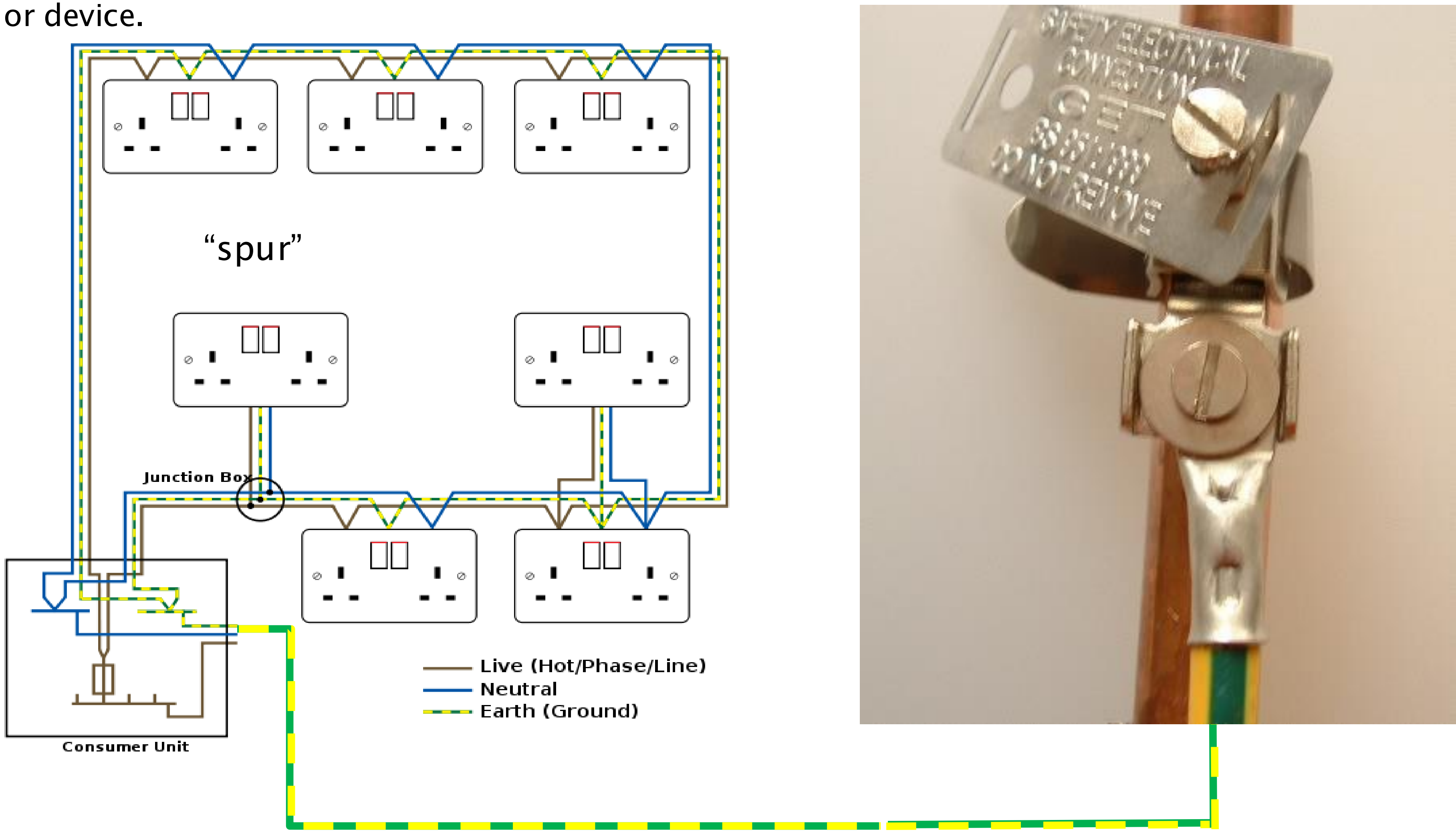


Live bus bar distributes the incoming live current to multiple circuit breakers.

House Ring Main Wiring

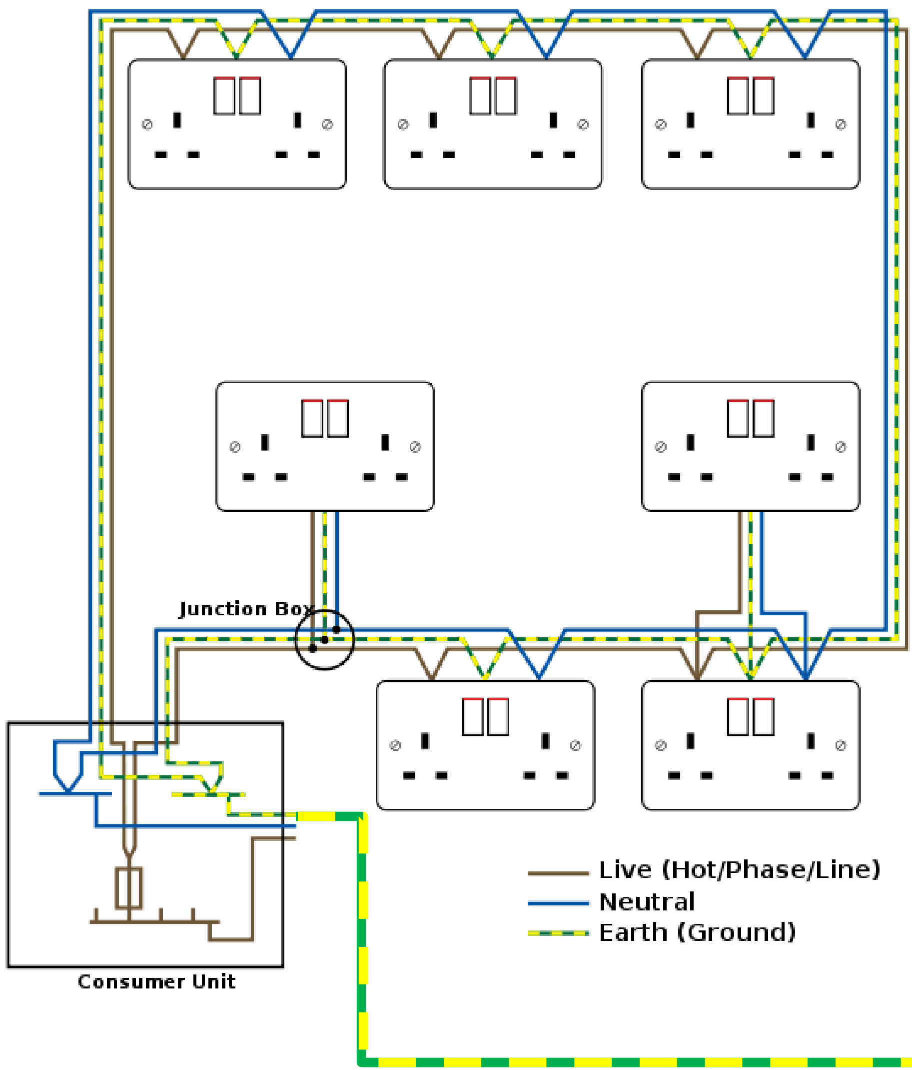
In a ring circuit, a spur is a branch that extends from a point on the ring (e.g., a socket or junction box) to supply an additional socket or device.

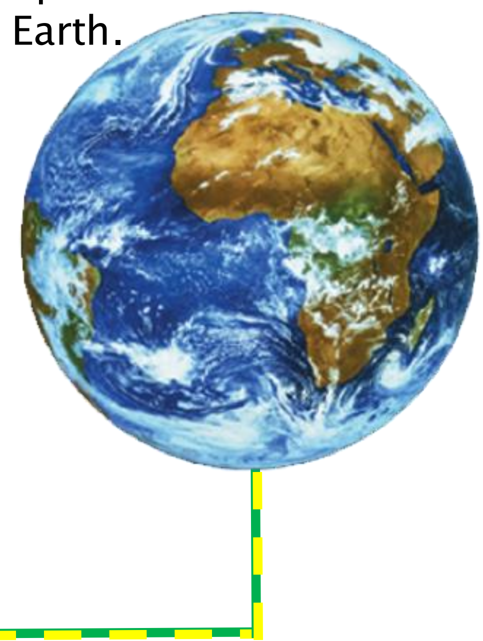
Earth is locally connected to suitable earth point, typically a water pipe, or an earth spike



House Ring Main Wiring

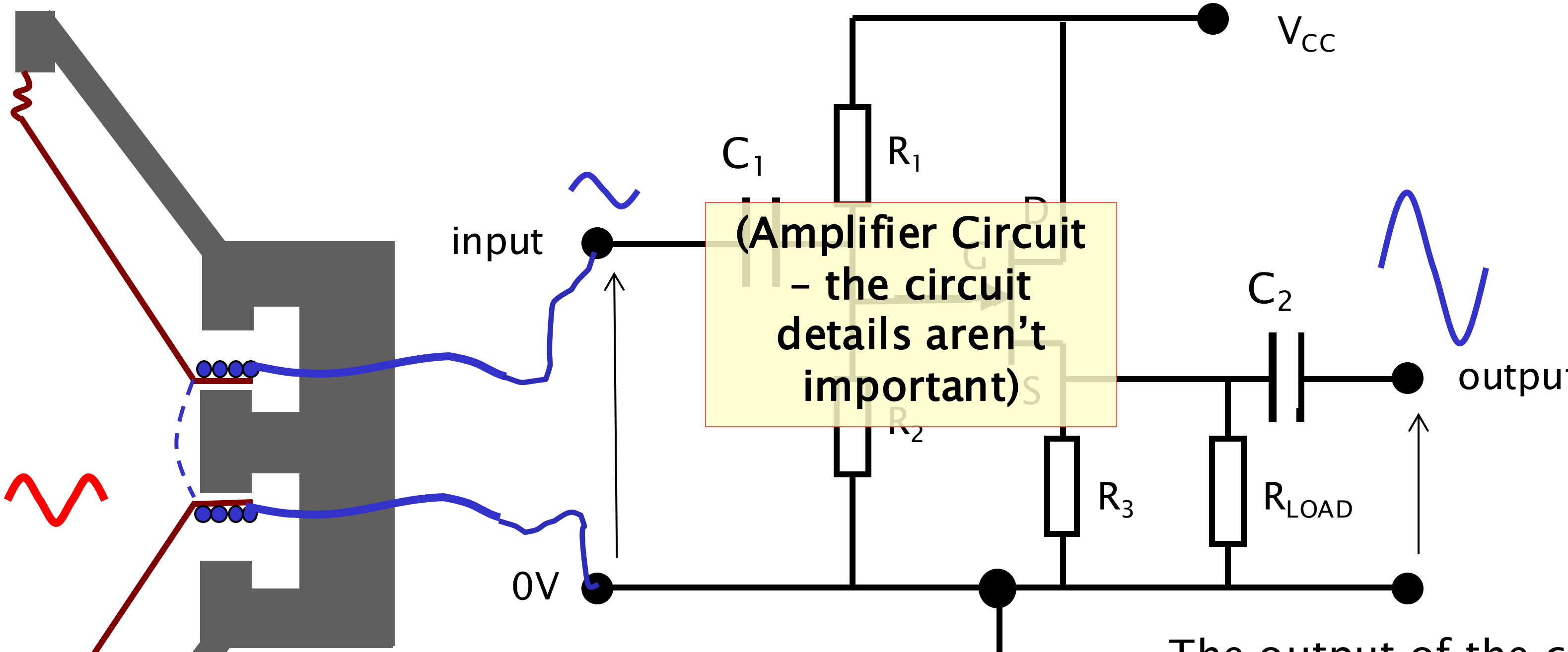
- Earth is a massive conductor and serves as a central reference point for electrical potential (assigned 0V by convention).
- The earth cable ensures that all metal parts of the electrical system (e.g., socket casings, device frames) are at the same potential as the Earth (0V), preventing electric shocks.
- During a fault (e.g., a live wire touching a metal casing), the earth cable provides a low-resistance path for the fault current to flow to the Earth.





Circuits – Input and Outputs

The amplifier boosts the weak signal to a level suitable for further processing (e.g., to drive a speaker or for recording).

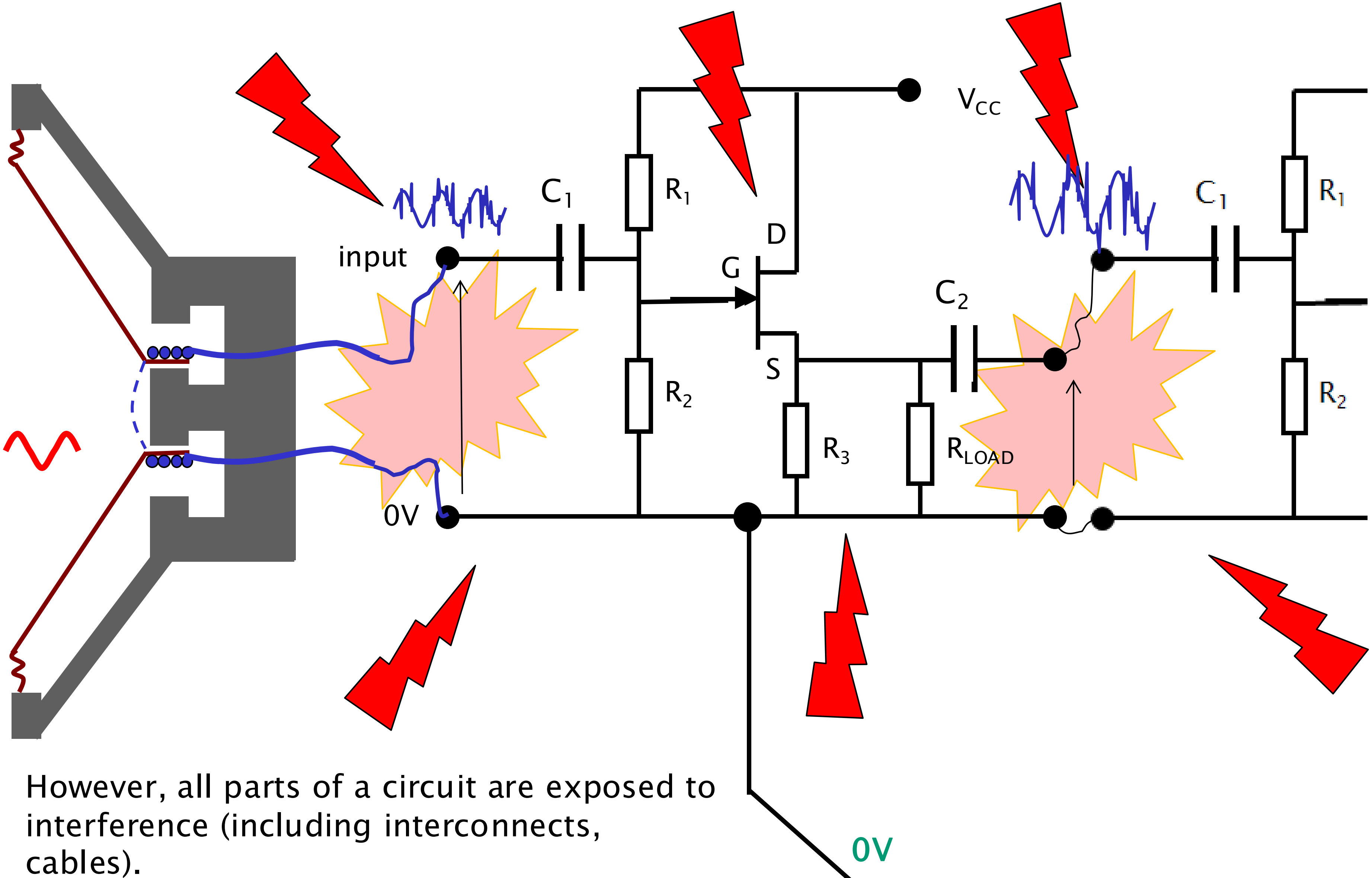


If one side of the mic coil is held at 0V, the other side of the coil is at whatever voltage is generated by the input sound; typically fraction of a volt

The output of the circuit changes with the input signal. If one terminal is held at 0V, the output varies by the amplification of the circuit (i.e. input voltage x Gain)

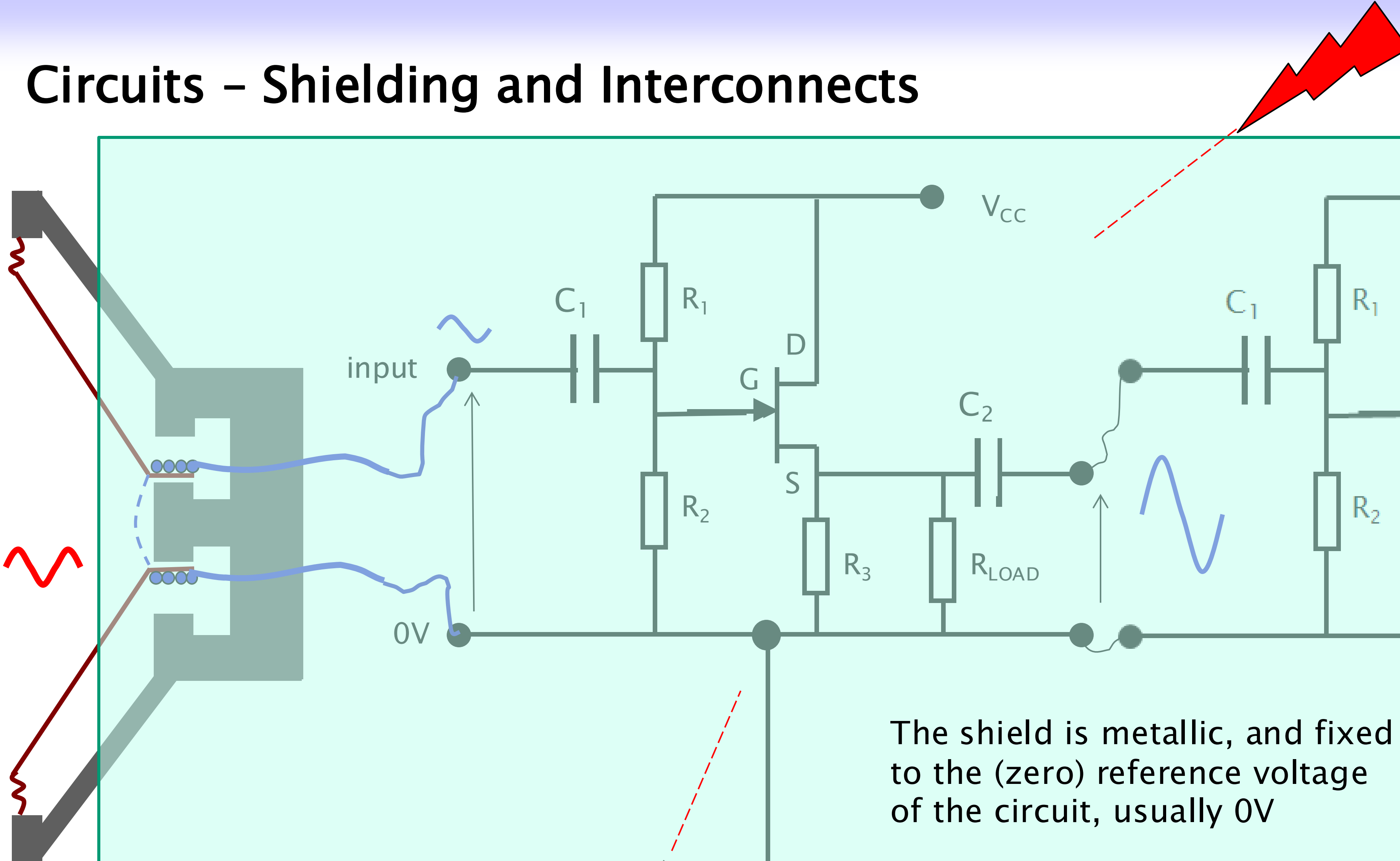
Zero Volts (could be Earth, maybe not)

Circuits – Interference



However, all parts of a circuit are exposed to interference (including interconnects, cables).
Interference can introduce noise into the audio signal.

Circuits – Shielding and Interconnects



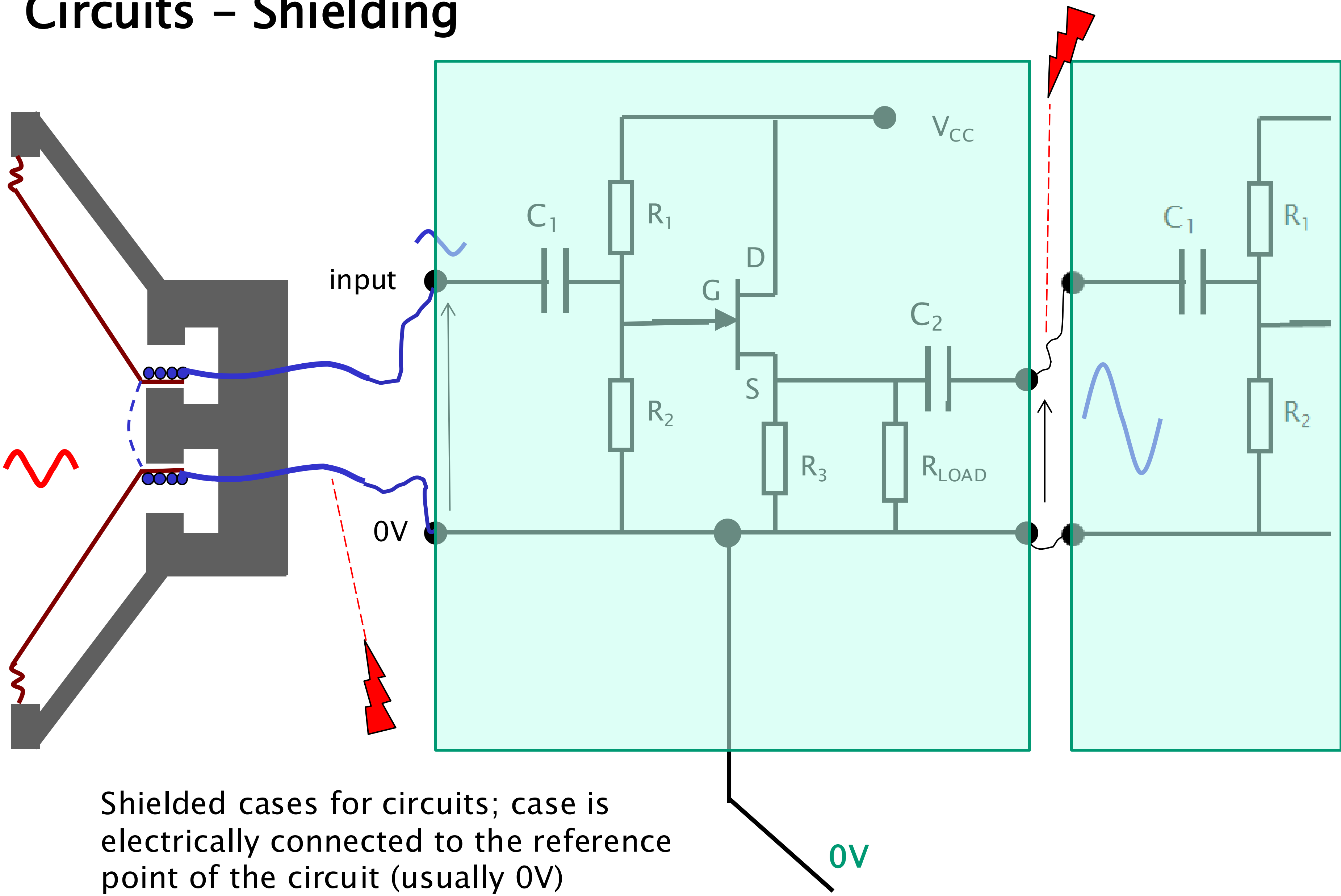
The shield is metallic, and fixed to the (zero) reference voltage of the circuit, usually 0V

Shielding is a fundamental technique in audio engineering to protect circuits and interconnects from EMI and radio frequency interference (RFI).

0V

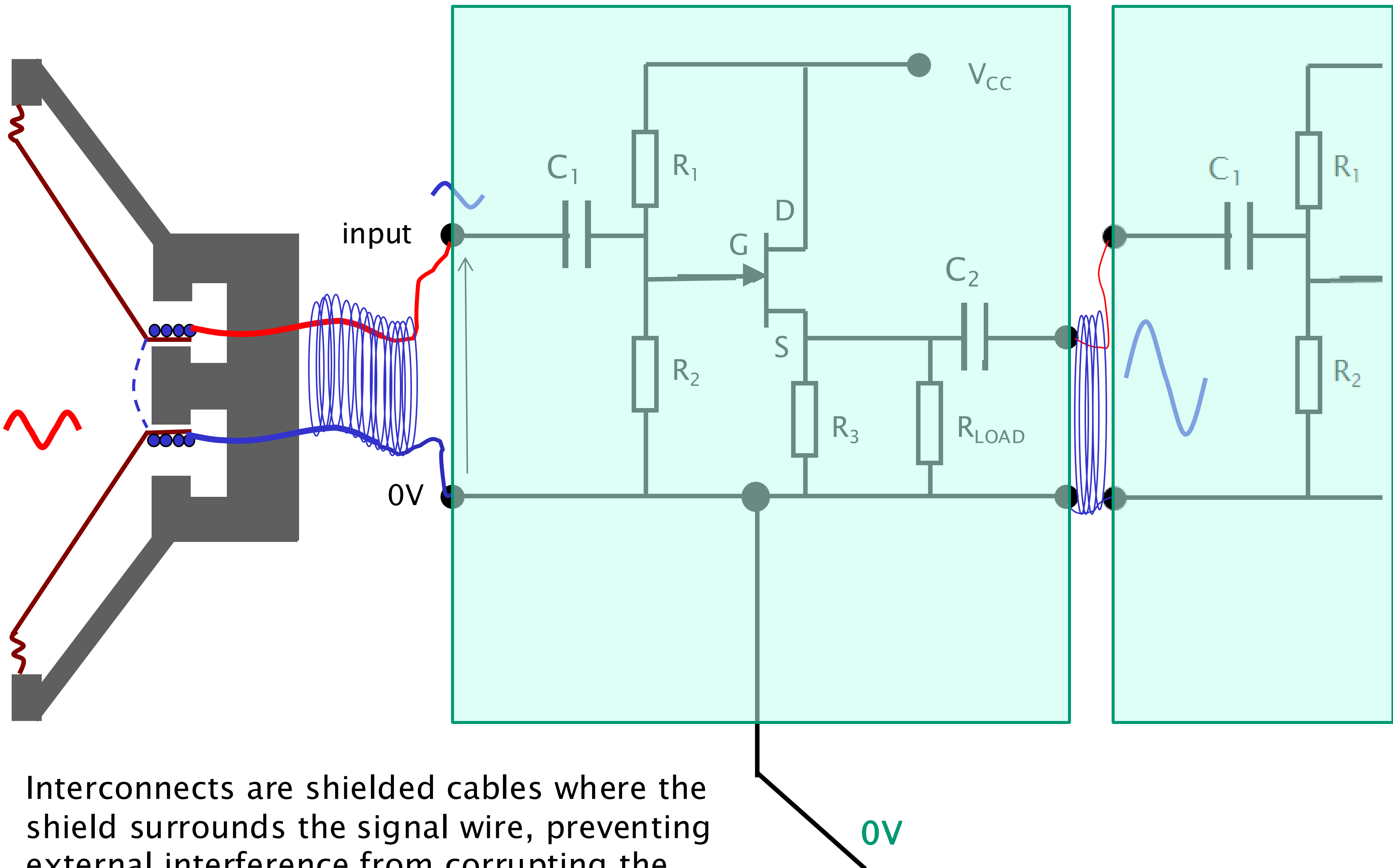
Circuits can be easily housed in metal cases, but cables present a problem (due to their required flexibility) …

Circuits – Shielding



Shielded cases for circuits; case is electrically connected to the reference point of the circuit (usually 0V)

Circuits – Shielding Interconnects

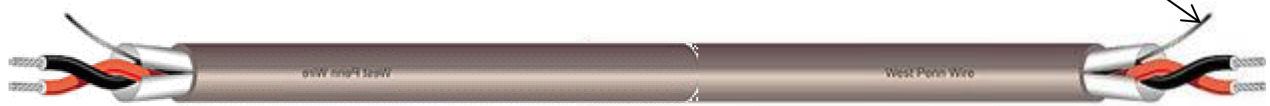


Interconnects are shielded cables where the shield surrounds the signal wire, preventing external interference from corrupting the signal.

Circuits – Shielding Interconnects

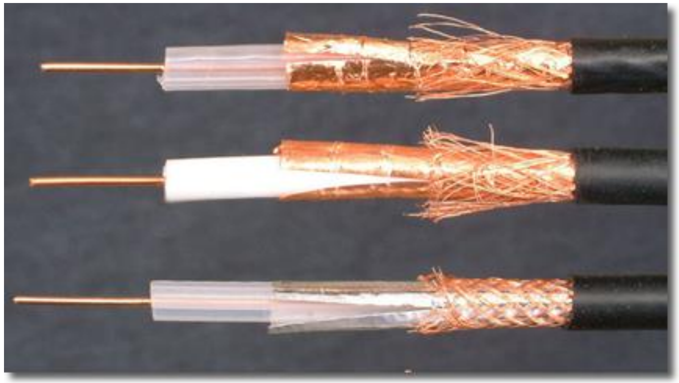
One or more signal wires

metallic shield (often a mesh) surrounding the signals, connected at plug via a wire running the length of the cable



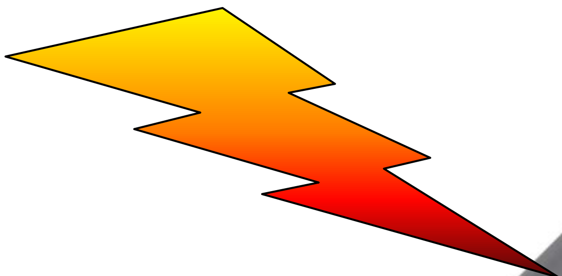
This still allows signal wires to interfere with each other, so the signals may be individually shielded

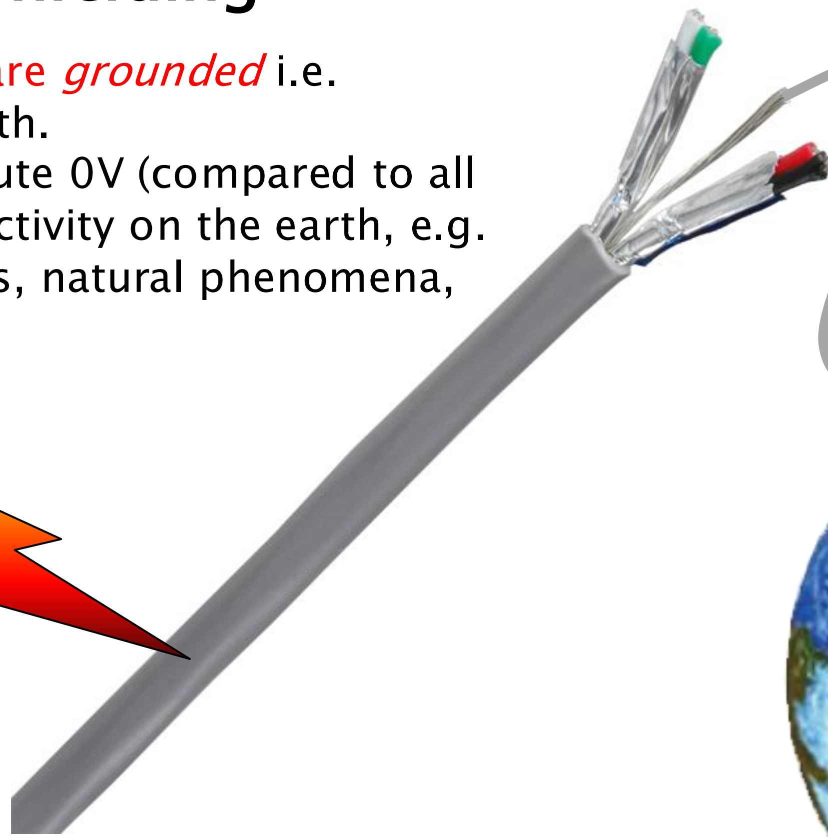
For radio frequencies, coax cable is required – shield has special resistive properties (see later)



Grounded Shielding

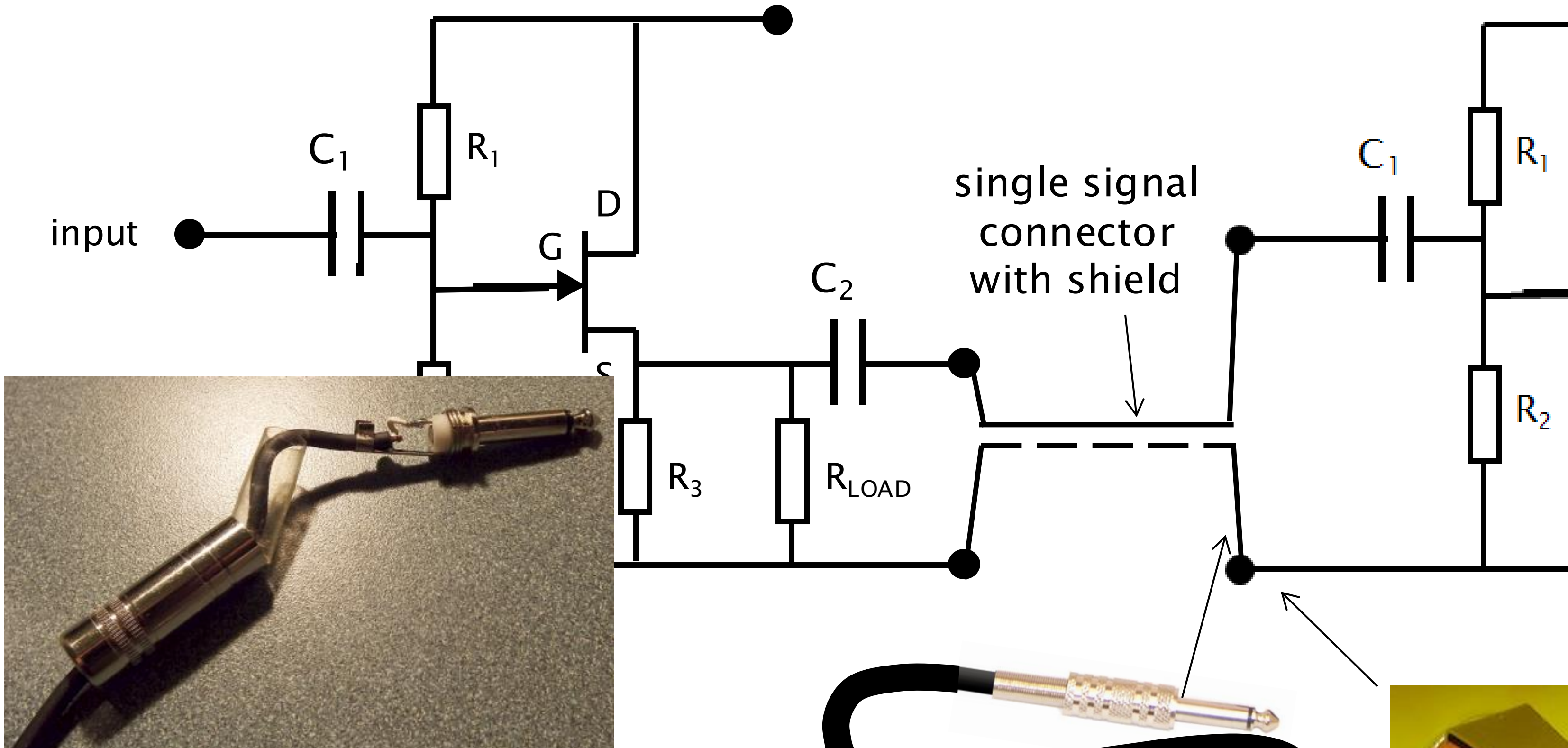
Usually, shields are *grounded* i.e. connected to Earth.
Earth is an absolute 0V (compared to all other electrical activity on the earth, e.g. electronic circuits, natural phenomena, lightening, etc)





- An earth shield provides protection
- Shown here – a four signal lead with individual shielding

Circuit Interconnection – Unbalanced



In an unbalanced connection:

•One conductor carries the audio signal (the tip of the plug and jack, the centre conductor of the coaxial cable).

•The other conductor (the sleeve of the plug and jack, the shield of the coaxial cable) serves as the ground return path for the signal and also provides shielding against EMI.

Other Plugs for Unbalanced Connections

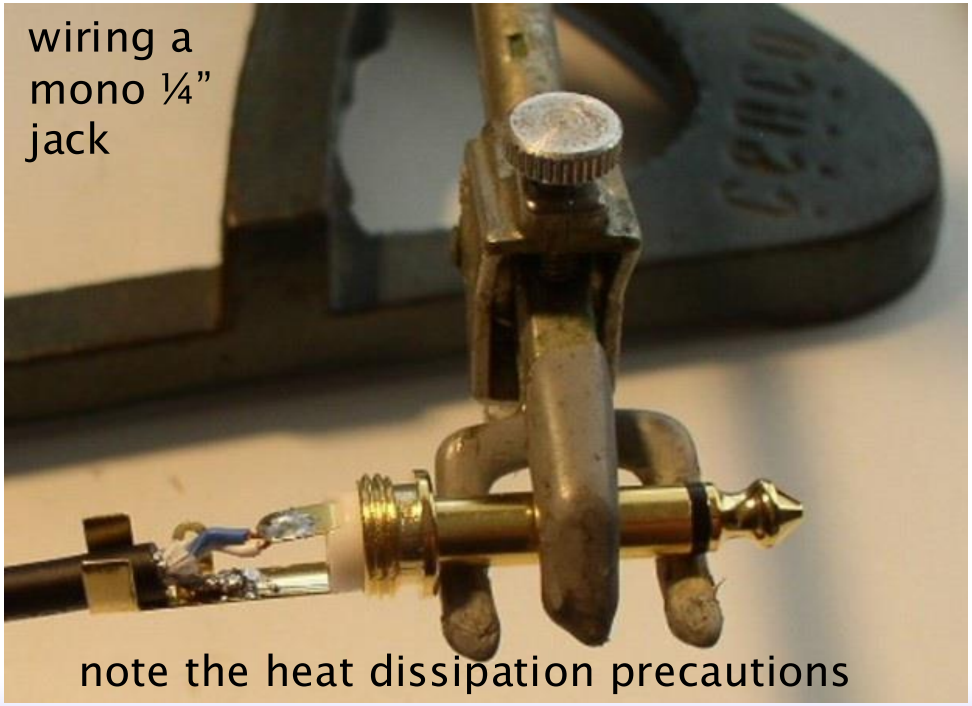
mono 3.5mm jack

5 pin DIN plug

stereo ¼” jack

stereo 3.5mm jack

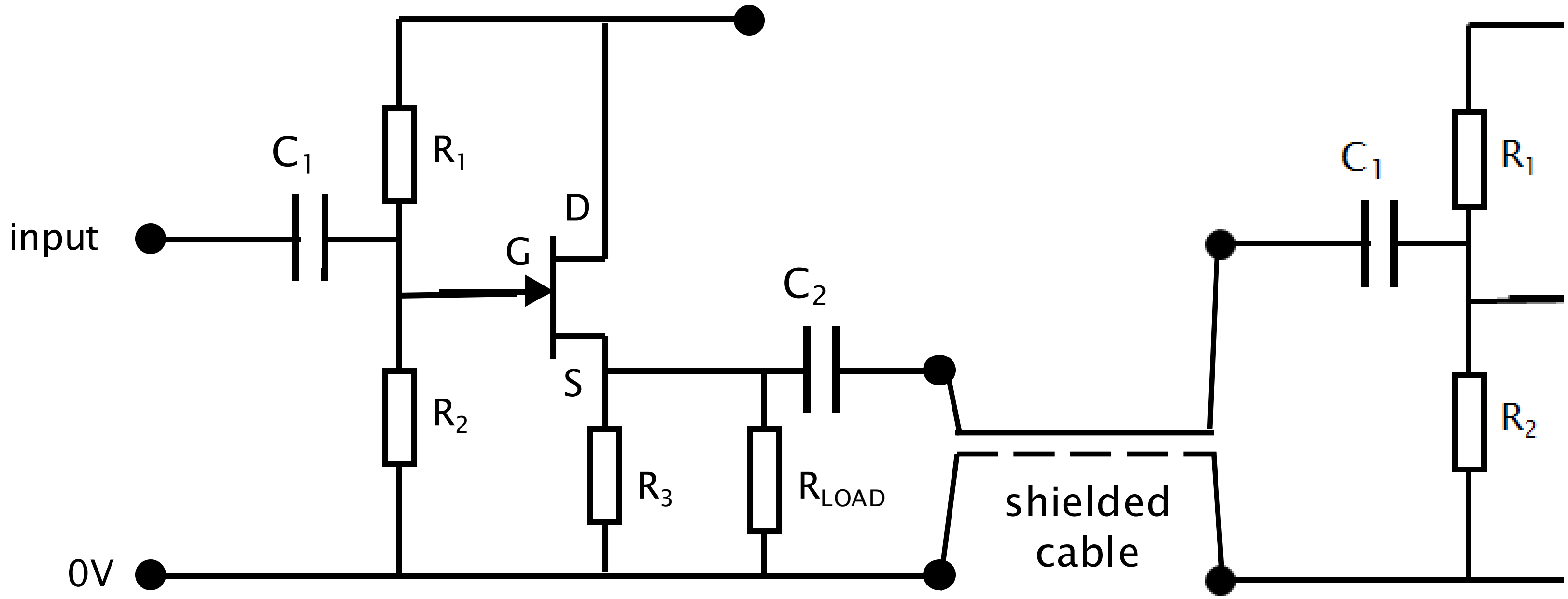
two mono phono jacks (for stereo)



wiring a mono ¼” jack

note the heat dissipation precautions

Circuits – Unbalanced Connection

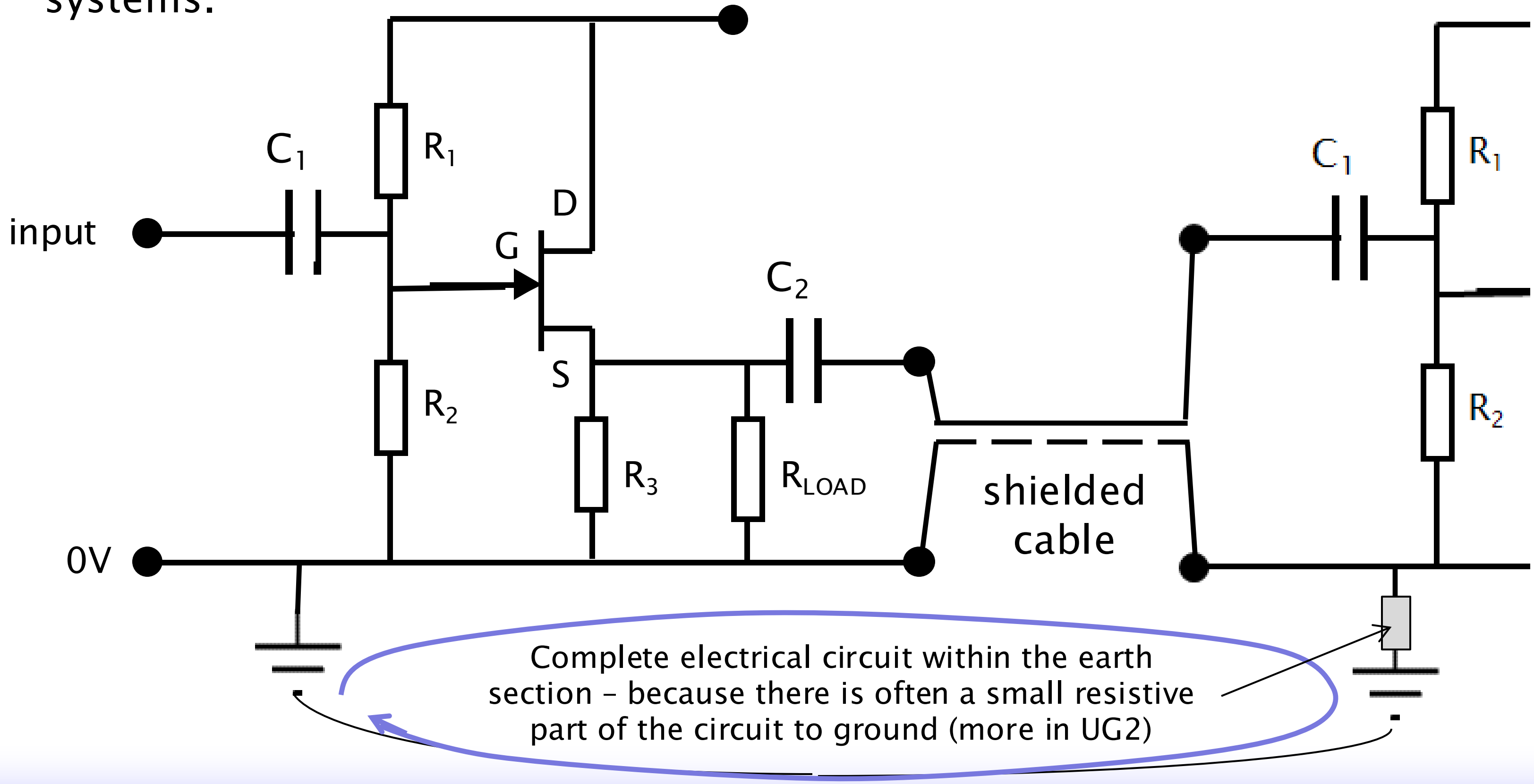


These interconnects (shown up to here) are <u>unbalanced</u>
- i.e. signal on one lead; the return path is via the shield / 0V / earth lead

Prone to noise, especially as length increases
- This is sufficient for domestic audio equipment (short distances, less demanding than professional applications)
- For longer signal runs (stage work, event recordings, broadcast, etc) – need more robust signal protection ...

Ground Loops

- Loops occur when there are multiple paths connecting the ground references of two interconnected pieces of equipment.
- If there is a difference in electrical potential between the "earth ground" points of the two units, a current will flow through this ground loop.
- This current flows through the shield of the cable can induce a noise voltage into the signal.
- Ground loops are a common source of unwanted noise in audio and electronic systems.



Ground Loops – Ground Lift

- **GND LIFT Switch** is a common control found on professional audio gear designed to help eliminate ground loop.
- It works by breaking the direct connection between the chassis ground and the signal ground at that specific input or output (disconnect (or "lift") the ground connection between two pieces of equipment).

should be Ok on professional equipment
… but always beware the safety aspects!!!

Professional Audio Signal Interconnects

Via a balanced line

- two signal conductors
- plus earth shielding

Uses circuitry to convert ...

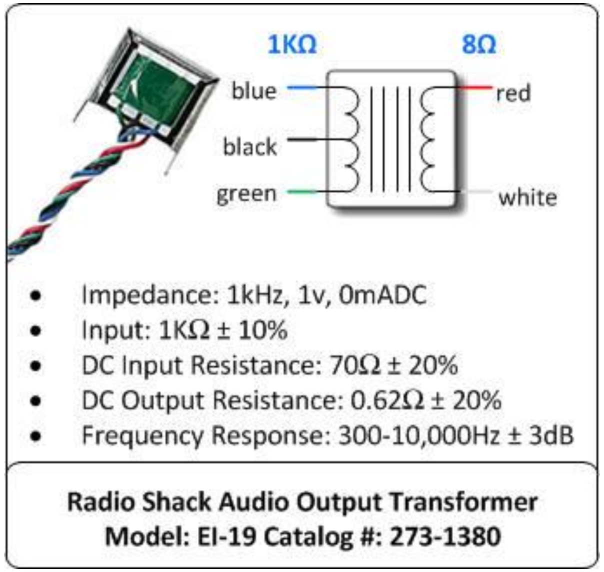
single-ended (unbalanced) line → balanced line → input to circuit:

Incoming audio signal is initially in an unbalanced format (signal on one conductor, ground on another) and needs to be converted to a balanced format before being processed by the circuit.

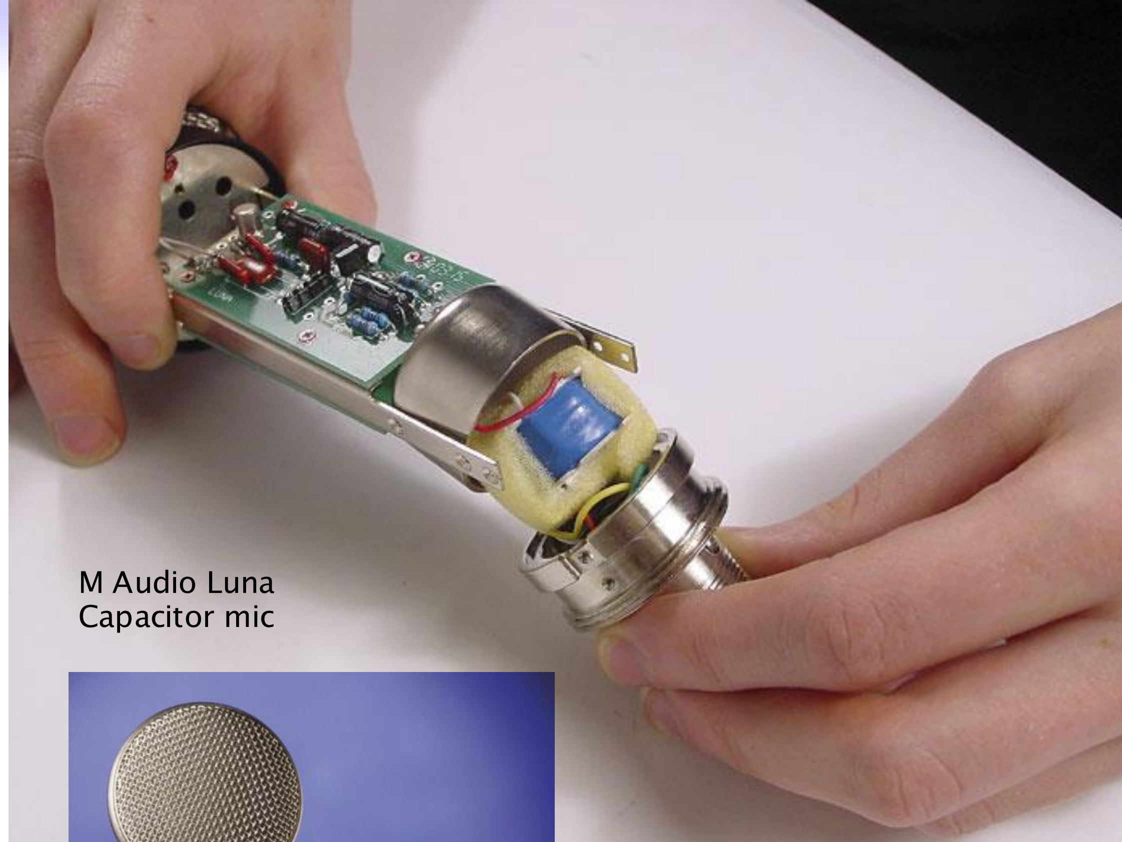
... achieved by

- transformer circuit
  - Passive (it does not require an external power supply) – no amplifier – no induced noise
  - op-amp (electronic circuit)
    - While op-amps can effectively perform the unbalanced-to-balanced conversion, the active electronics themselves can potentially introduce some noise into the signal.
    - op-amp is an active device—it needs power to work and amplifies signals using transistors.
      ... (considering we are aiming to eliminate noise in the first place ...)

Audio Transformer

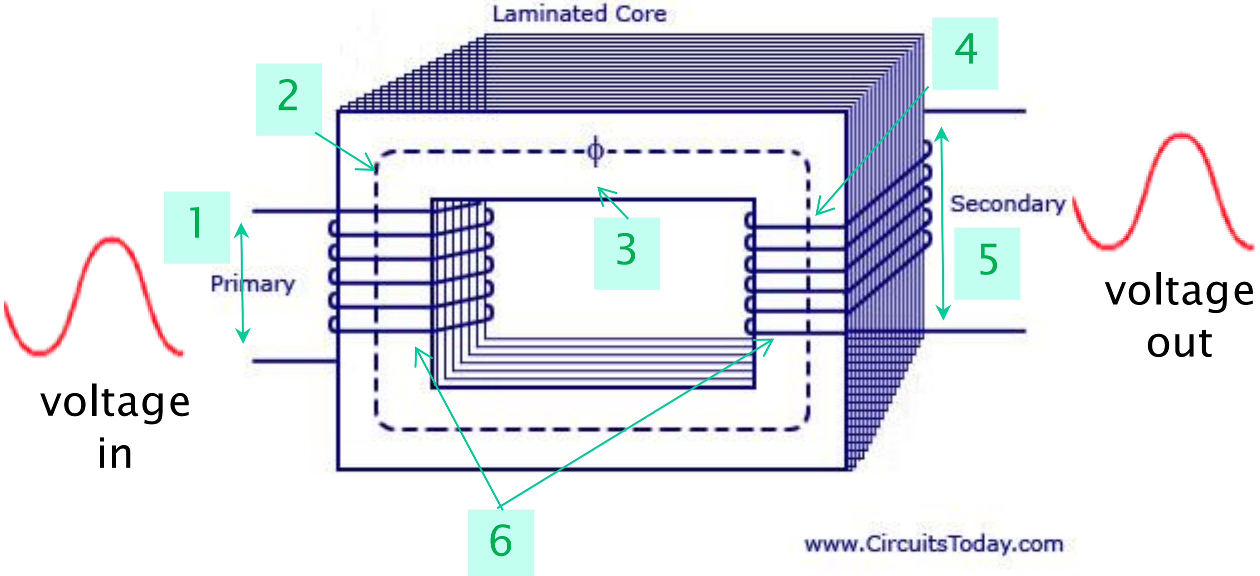


- The transformer takes a 1V audio signal from the mic (with 1KΩ impedance) and converts it to a lower-impedance signal (8Ω) for the output device, like a speaker.
- It is built for AC audio signals only, with no DC current, to ensure clean sound without distortion.



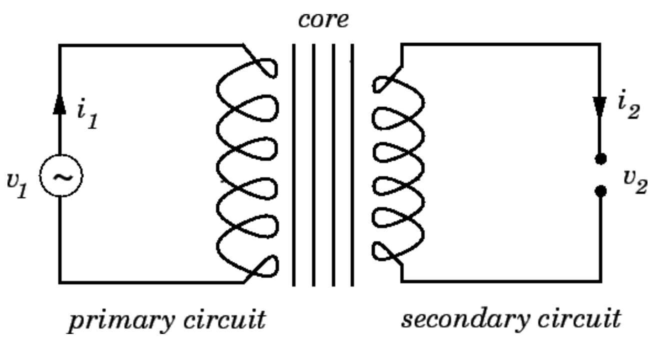
M Audio Luna Capacitor mic

Transformer

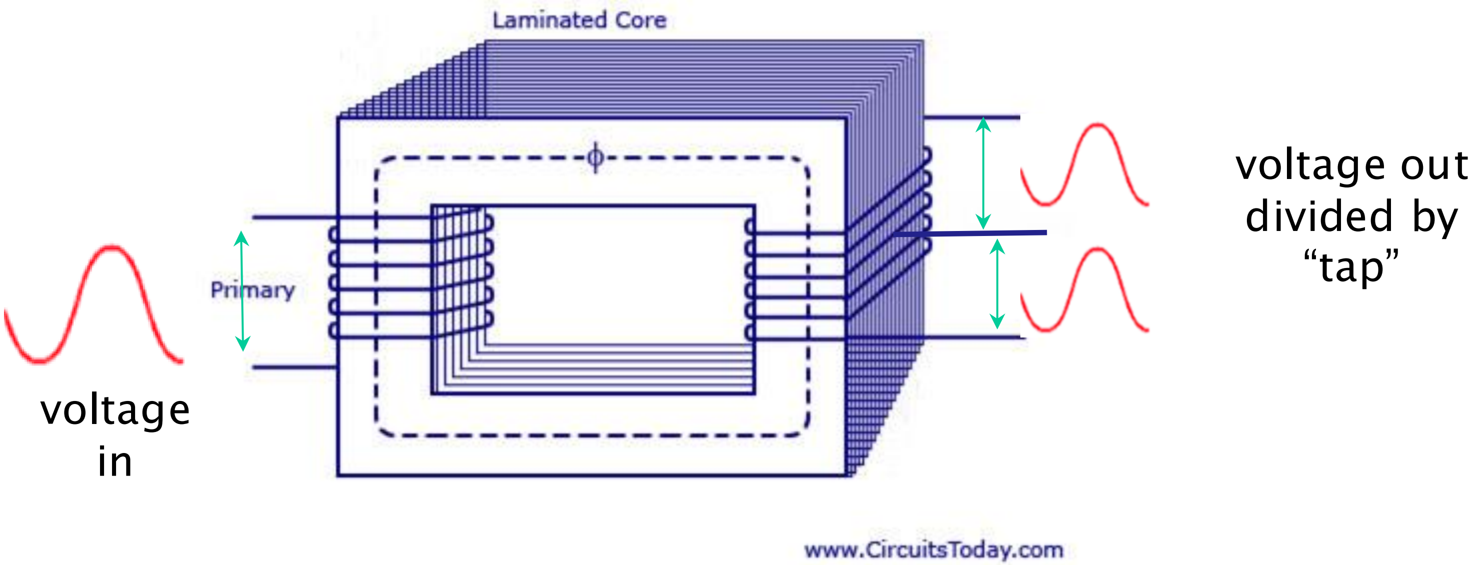


1. Voltage goes into primary winding
2. induces a magnetic field (AC signal creates a changing magnetic field in the transformer's core (the "Laminated Core"))
3. magnetic field strength is relative to the input signal
4. Changing magnetic field induces voltage in secondary winding
5. … voltage out
6. V_{out} depends on numbers of windings Primary:Secondary. (turns ratio)

Circuit schematic



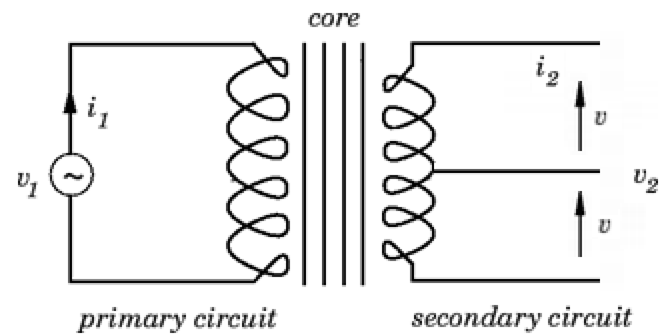
Transformer



A **centre-tapped transformer** is a type of transformer where the winding is split into two equal halves by a tap (connection point) at the midpoint of the winding.

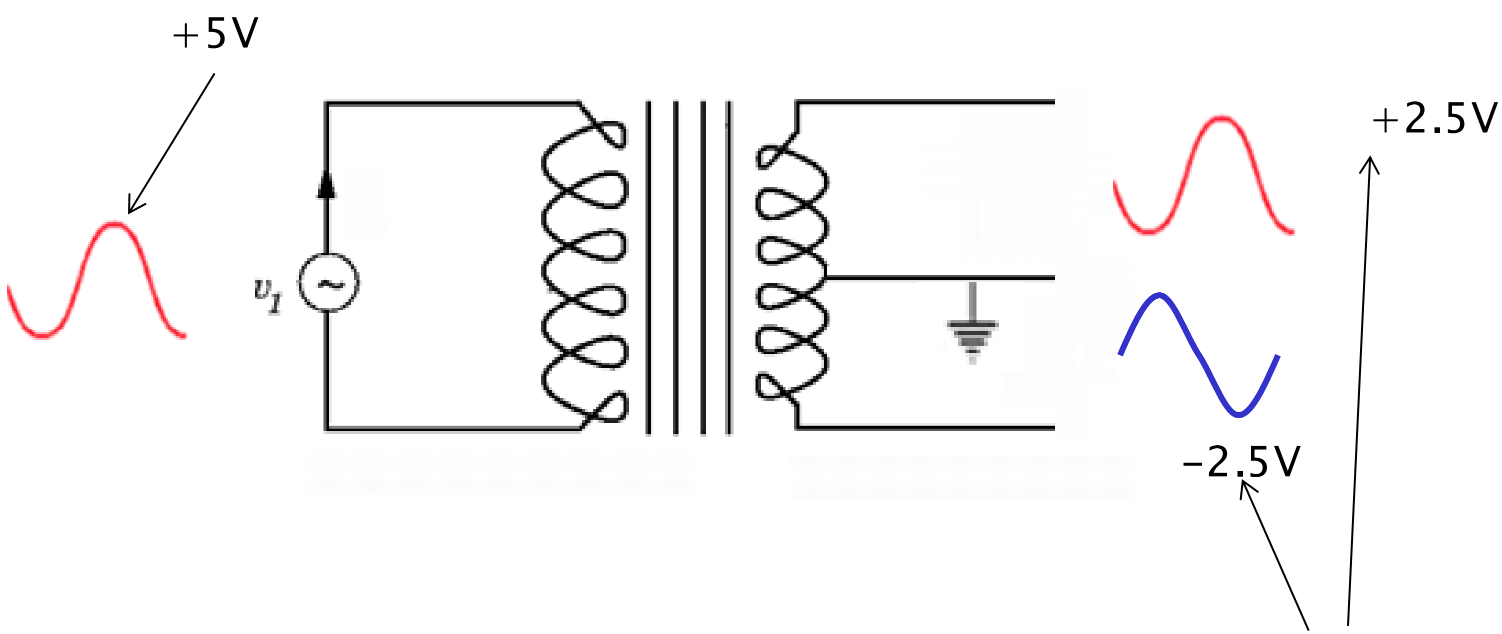
This centre tap divides the secondary winding into two sections, each producing a voltage that is half of the total secondary voltage relative to the centre tap.

Circuit schematic



Transformer – Centre Tapped, Earthed

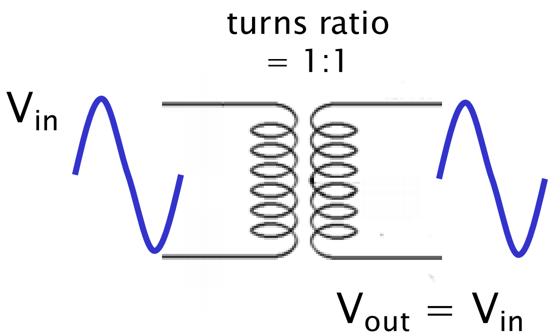
Input signal – at an instant when it reaches (e.g.) 5V …

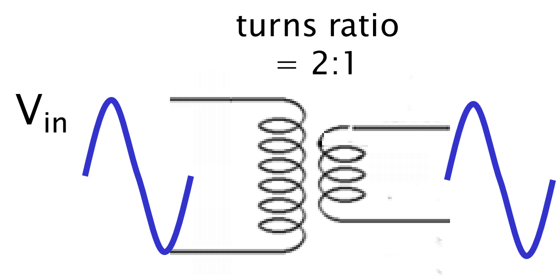


A **centre-tapped transformer** used to create two output signals that are **equal in amplitude but opposite in phase (out of phase)**, with the centre tap connected to **earth (ground).**

Output signal is divided to reach peaks of:
(i) +2.5V
(ii) –2.5V

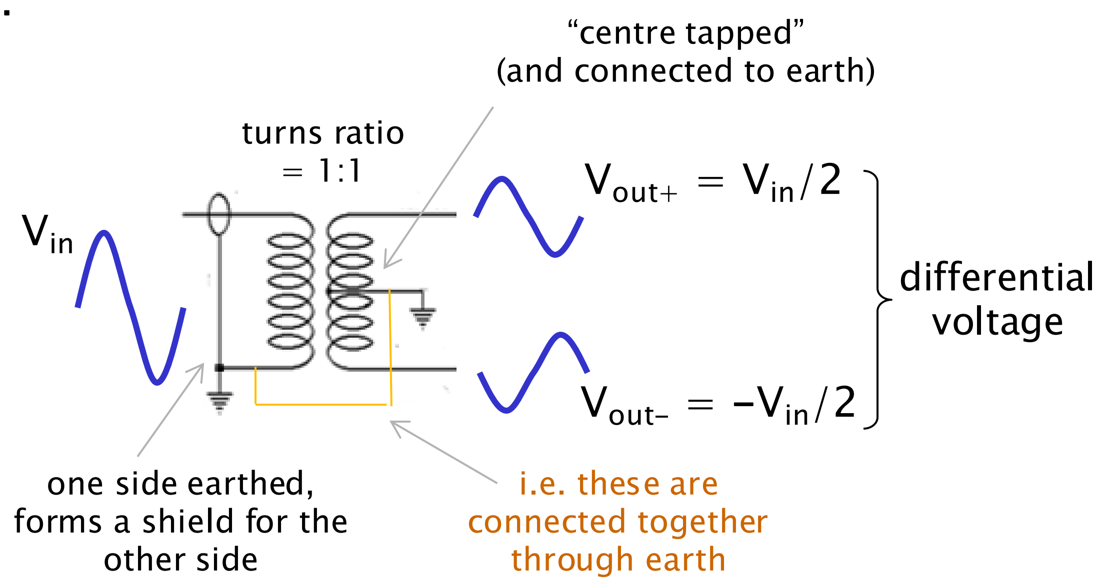
Transformer



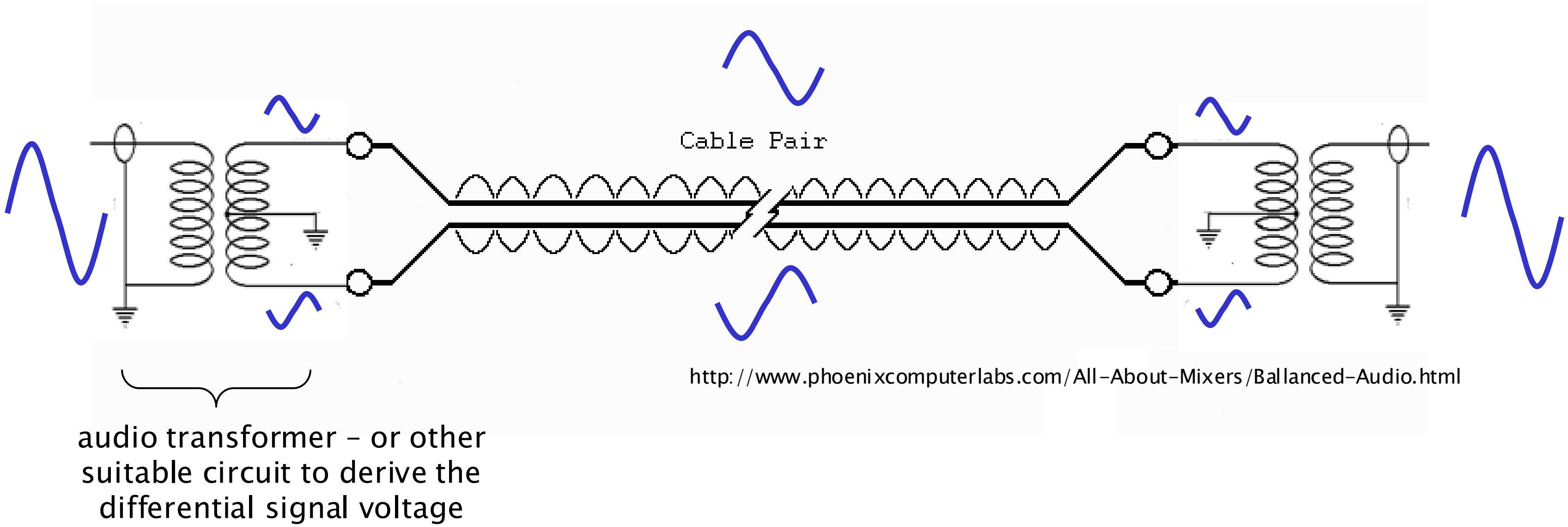


e.g. 240V in ... 120 V out

For an Audio Transformer:



Balanced Line – Transformer Circuit



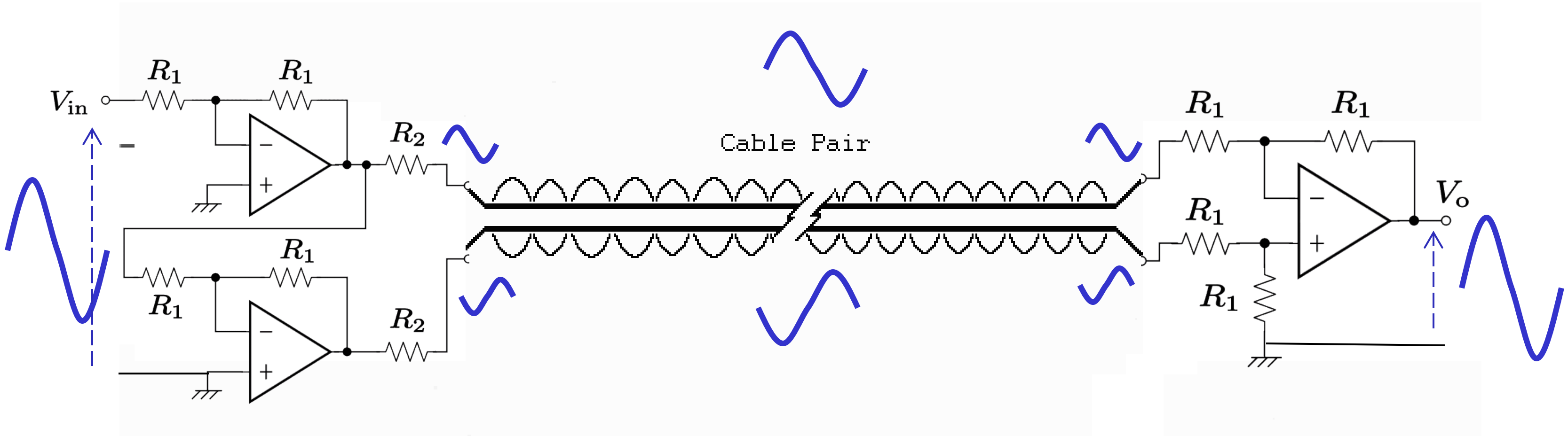
Unbalanced "single ended" source is converted to balanced signal

- e.g. via transformer

Signal is conveyed in a go/return normal/inverted circuit

- two lines are out of phase with each other
- ground noise (power line hum and buzz) and other interference imposed on one line is cancelled on the other
- transformer at load-end converts back to single ended
- cable signals are twisted together – further immunity to EM radiation
- signal conductors have equal impedances to ground

Balanced Line – Semiconductor Circuit



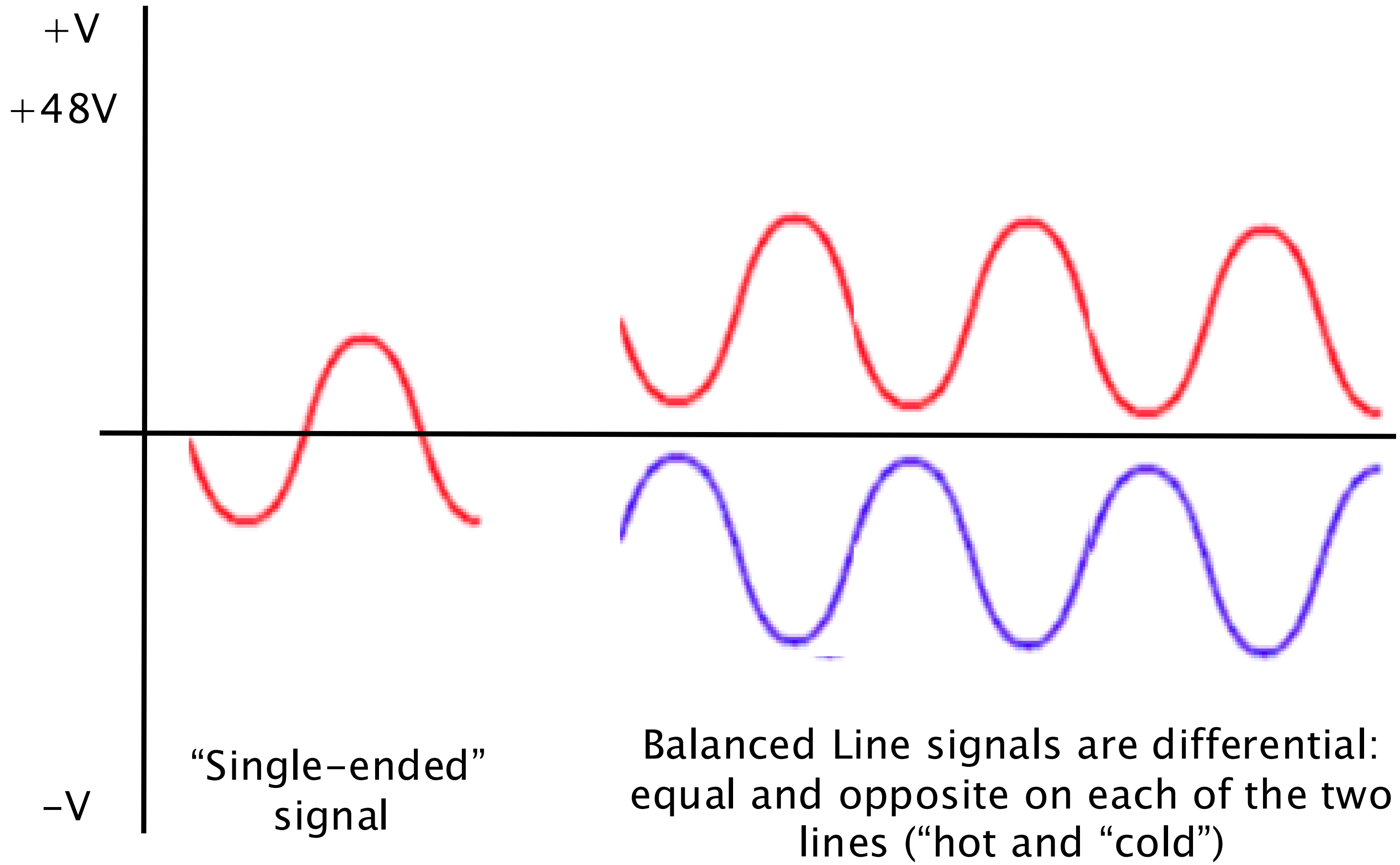
Unbalanced “single ended” source is converted to balanced signal

- e.g. via op-amp circuit

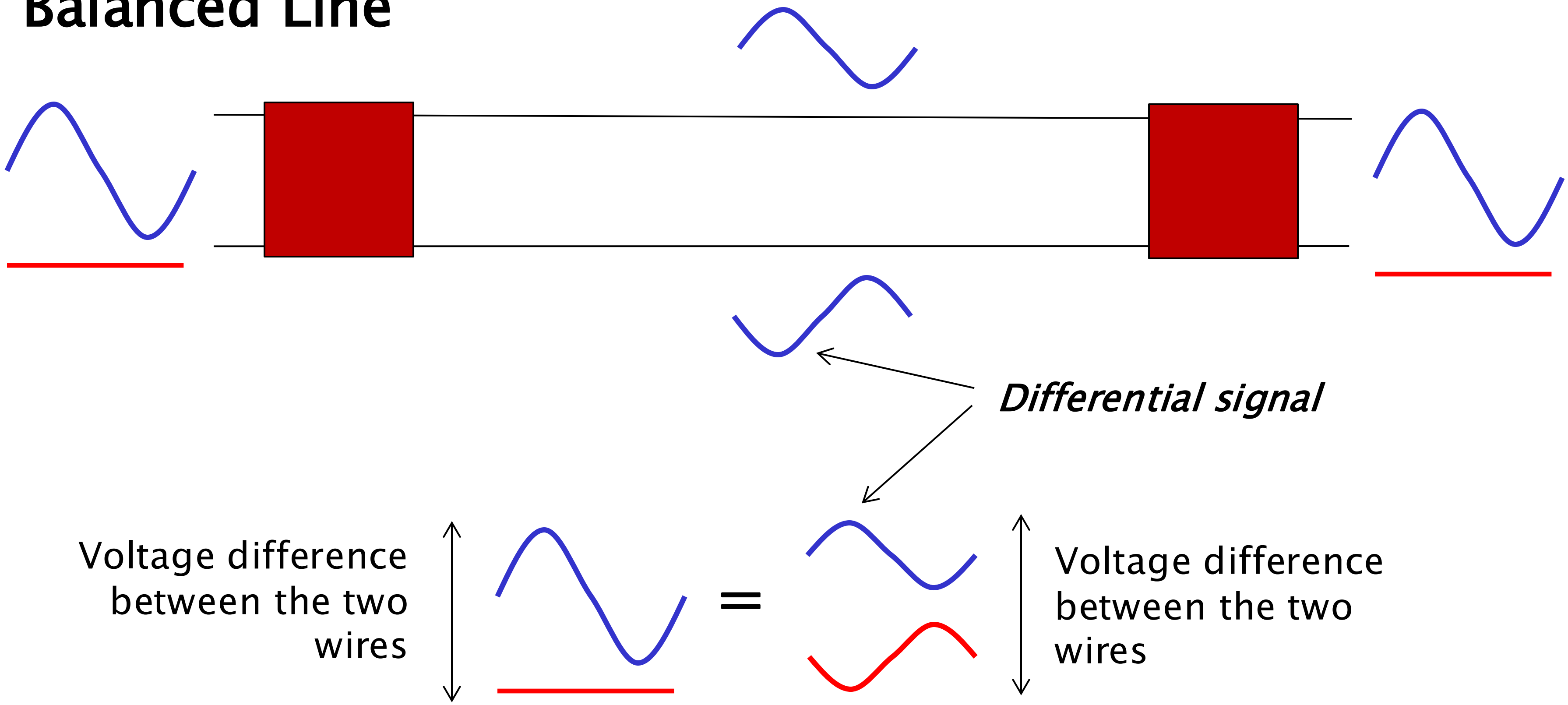
Signal is conveyed in a go/return normal/inverted circuit

- two lines are out of phase with each other
- ground noise (power line hum and buzz) and other interference imposed on one line is cancelled on the other
- op-amp circuit at load-end converts back to single ended
- cable signals are twisted together – further immunity to EM radiation
- signal conductors have equal impedances to ground

Balance Line Signal Voltages



Balanced Line

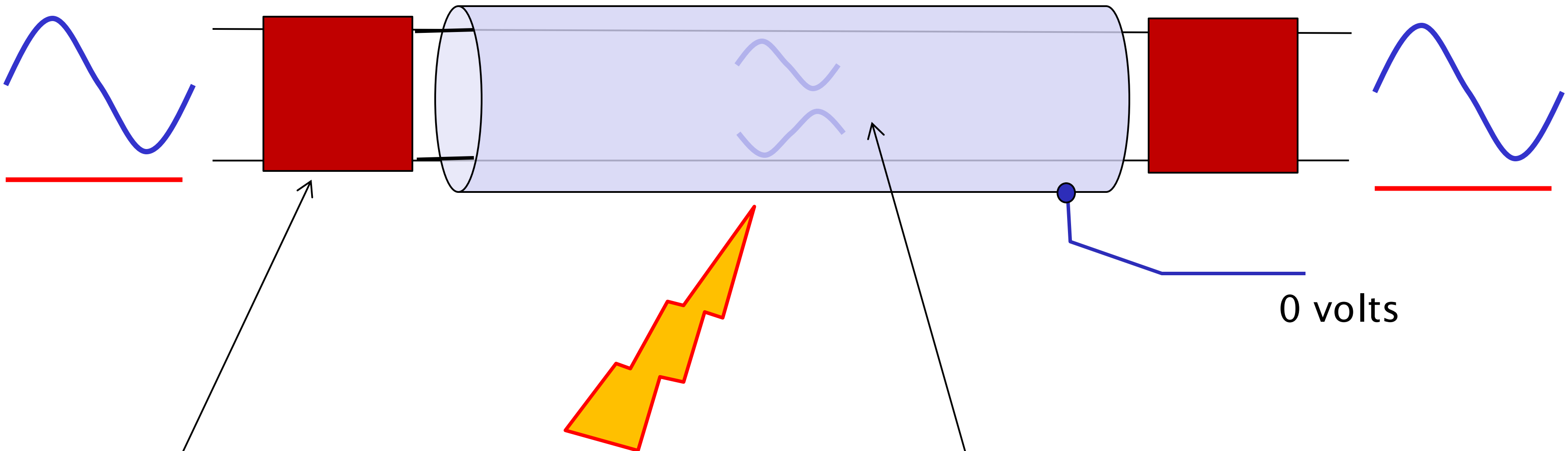


Differential Amplification: The receiving circuitry in a balanced system is designed to amplify the *difference* in voltage between the two signal lines.

Signal Preservation: Since the original signal is sent as two equal but opposite voltages, their difference is twice the amplitude of the individual signals (if referenced to ground).

Balanced Line

The signal is carried as a differential signal on two conductors
A third conductor forms the shield



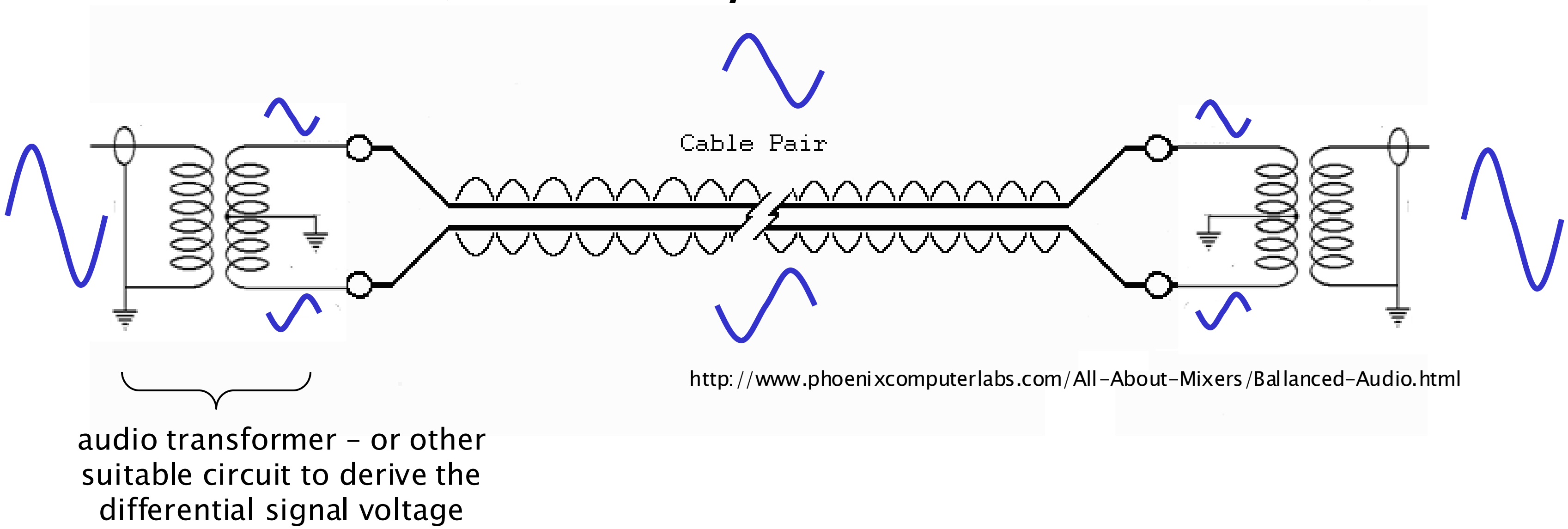
Shield provides noise immunity

Noise Cancellation: When noise is present on both lines, the differential amplifier subtracts the voltage on one line from the voltage on the other.

Since the noise is the same on both, it cancels out, leaving only the amplified original signal.

Audio transformer

Balanced Line *(will be explained further in UG2)*



Unbalanced "single ended" source is converted to balanced signal

- e.g. via transformer

Signal is conveyed in a go/return normal/inverted circuit

- two lines are out of phase with each other
- ground noise (power line hum and buzz) and other interference imposed on one line is cancelled on the other
- transformer at load-end converts back to single ended
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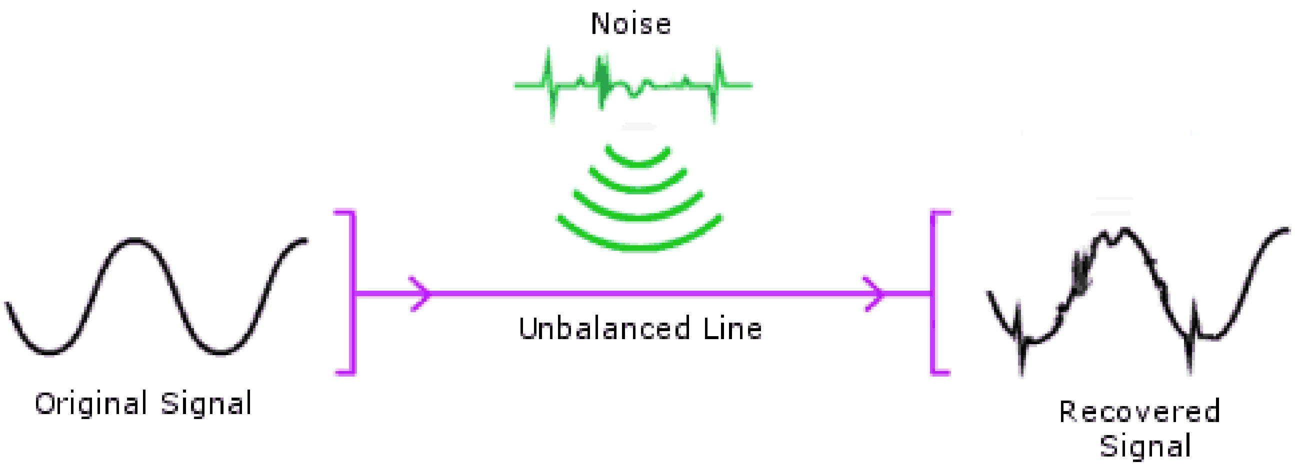
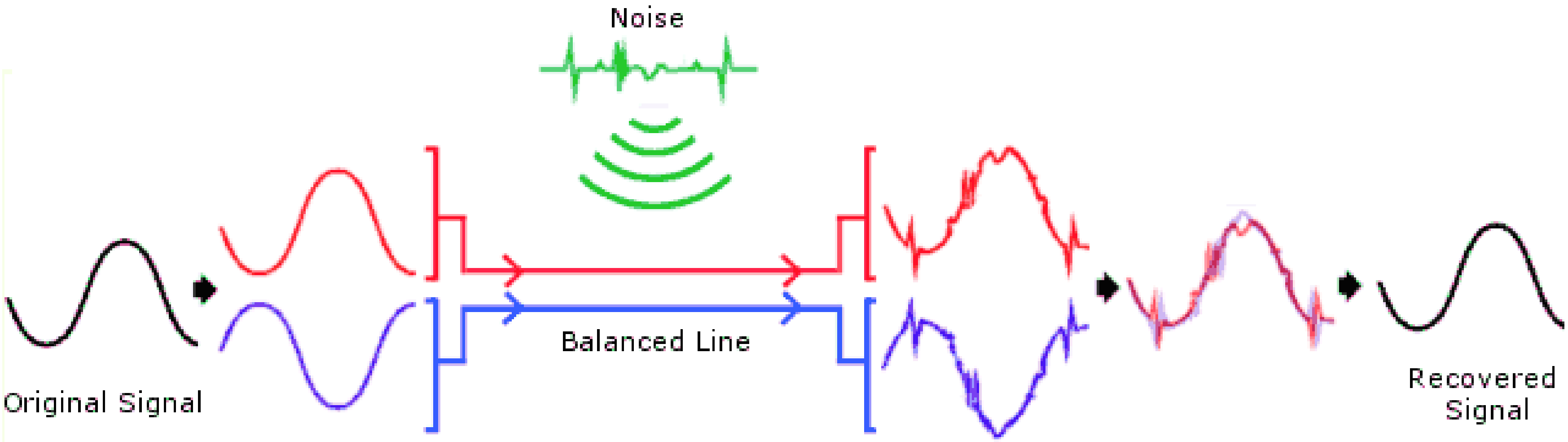
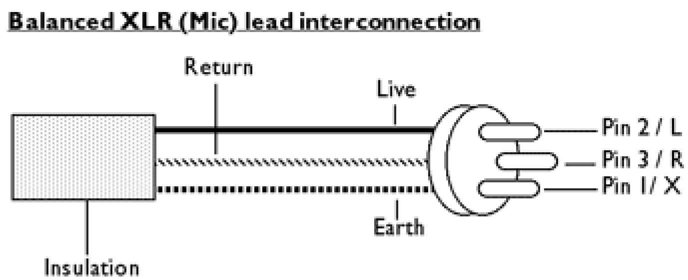
- signal conductors have equal impedances to ground

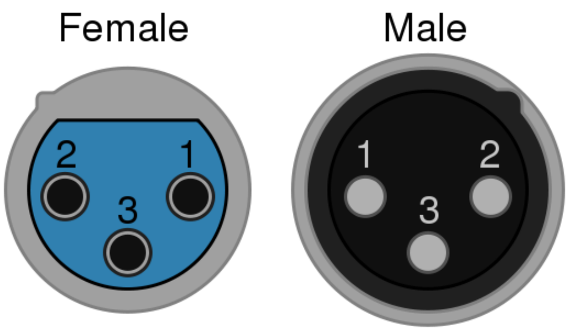
Balanced / Unbalanced – Comparison of Noise Performance





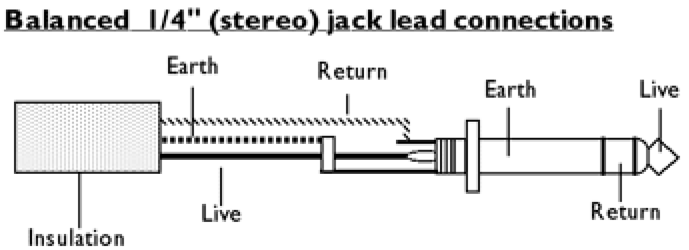
Balanced Line Connectors





Pin	Function
1	chassis ground (cable shield)
2	positive polarity terminal ("hot")
3	return terminal ("cold")

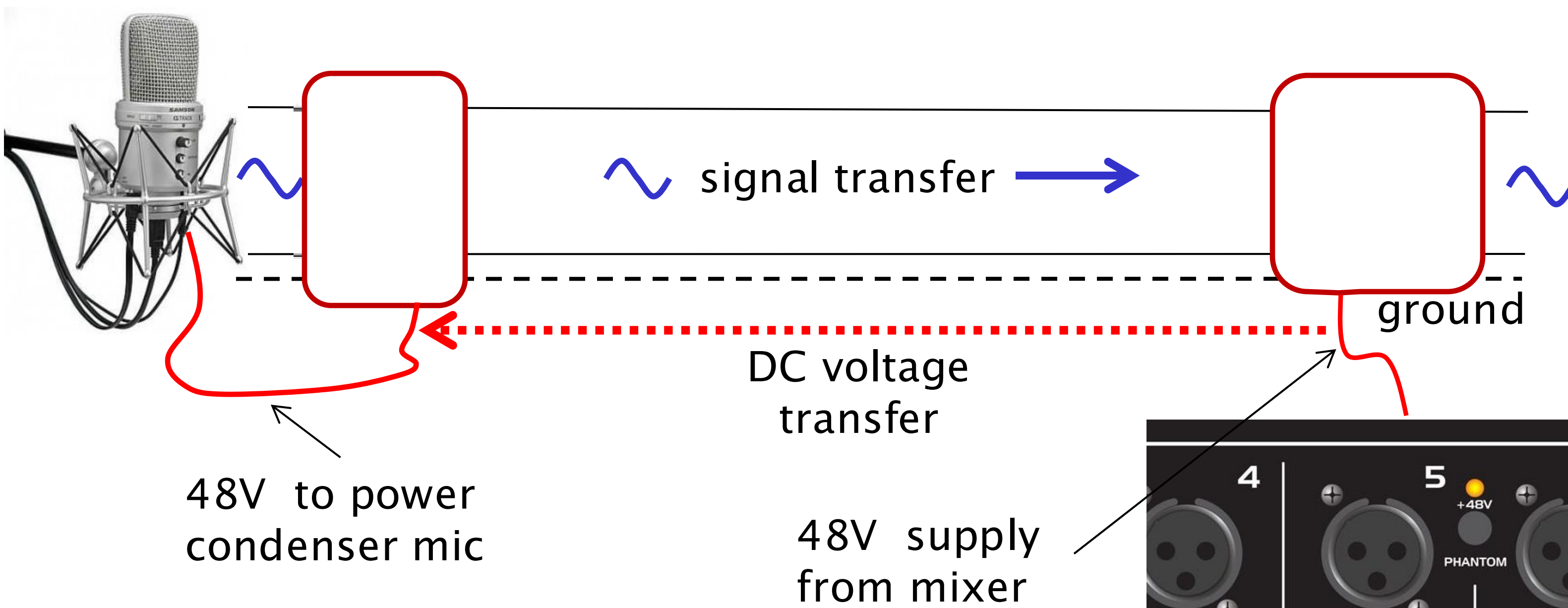
XLR Cables: Commonly used for microphones, mixers, and other professional audio equipment.

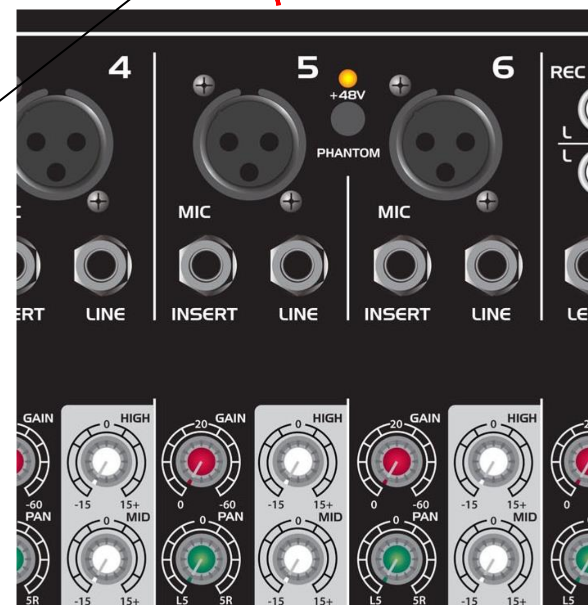


1/4" TRS Cables: Often used in studio environments for connecting instruments, audio interfaces, and monitors.

Phantom Power

Microphone signal and DC voltage supply are supported on the two connectors between the microphone and the mixer

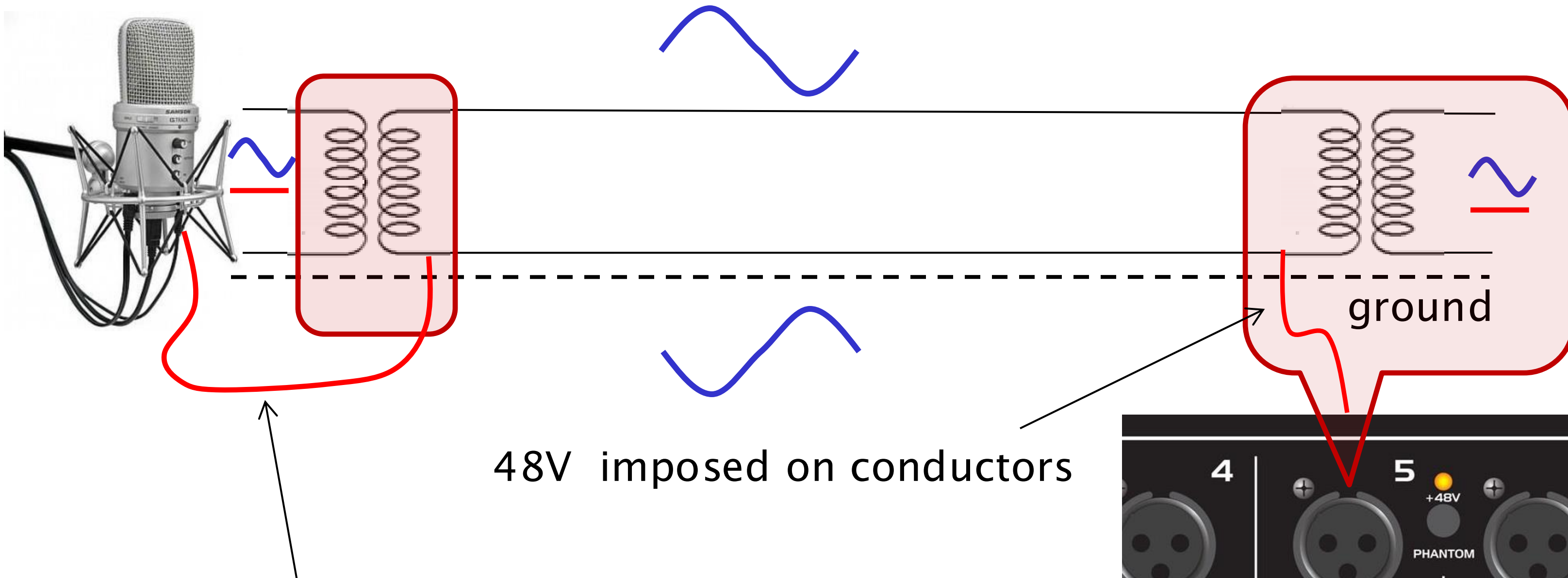


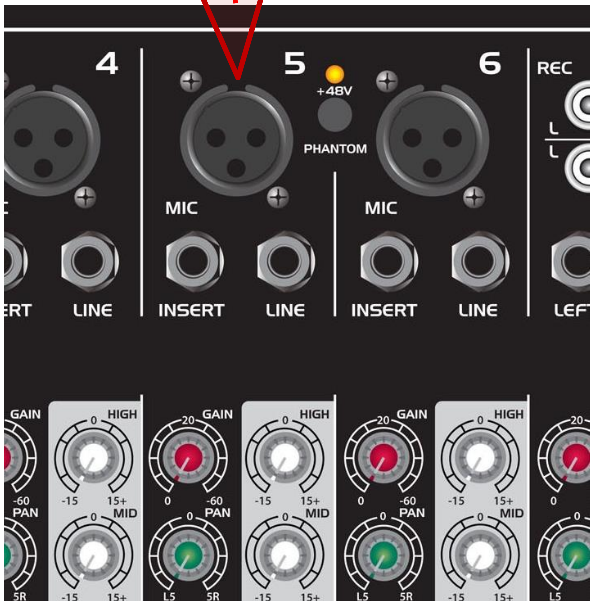


- Phantom power is a DC voltage (typically 48V) supplied by the mixing console to power condenser microphones.

- Condenser microphones require phantom power to operate their internal electronics, such as the preamplifier or impedance converter that boosts the signal from the microphone's capsule.

Phantom Power



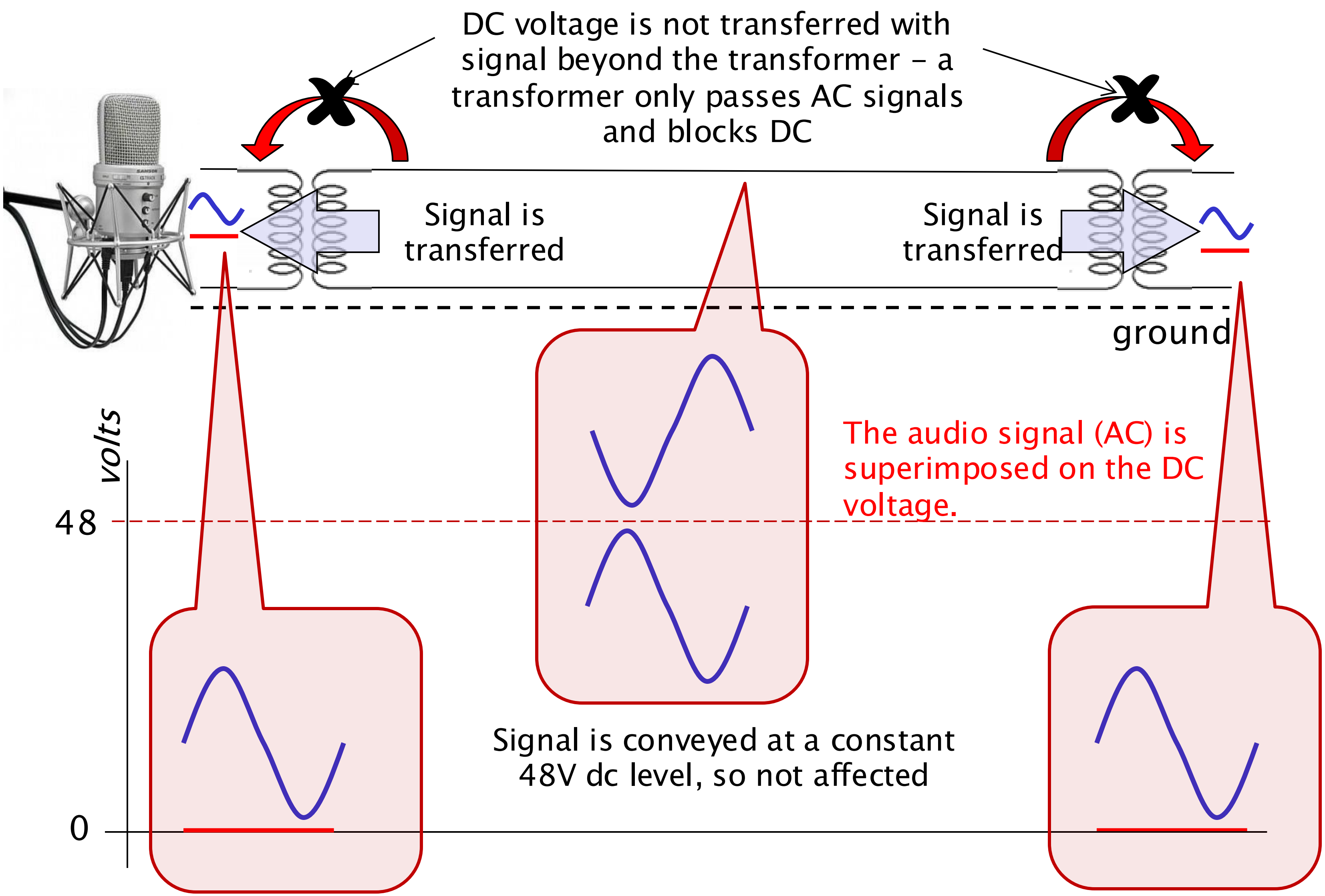


- +48V DC phantom power supplied by the audio mixer is reaching the microphone through the XLR cable.

- +48V DC is present at the microphone's connector and used for its operation, it is specifically **blocked or isolated** from the delicate audio signal path within the microphone itself.

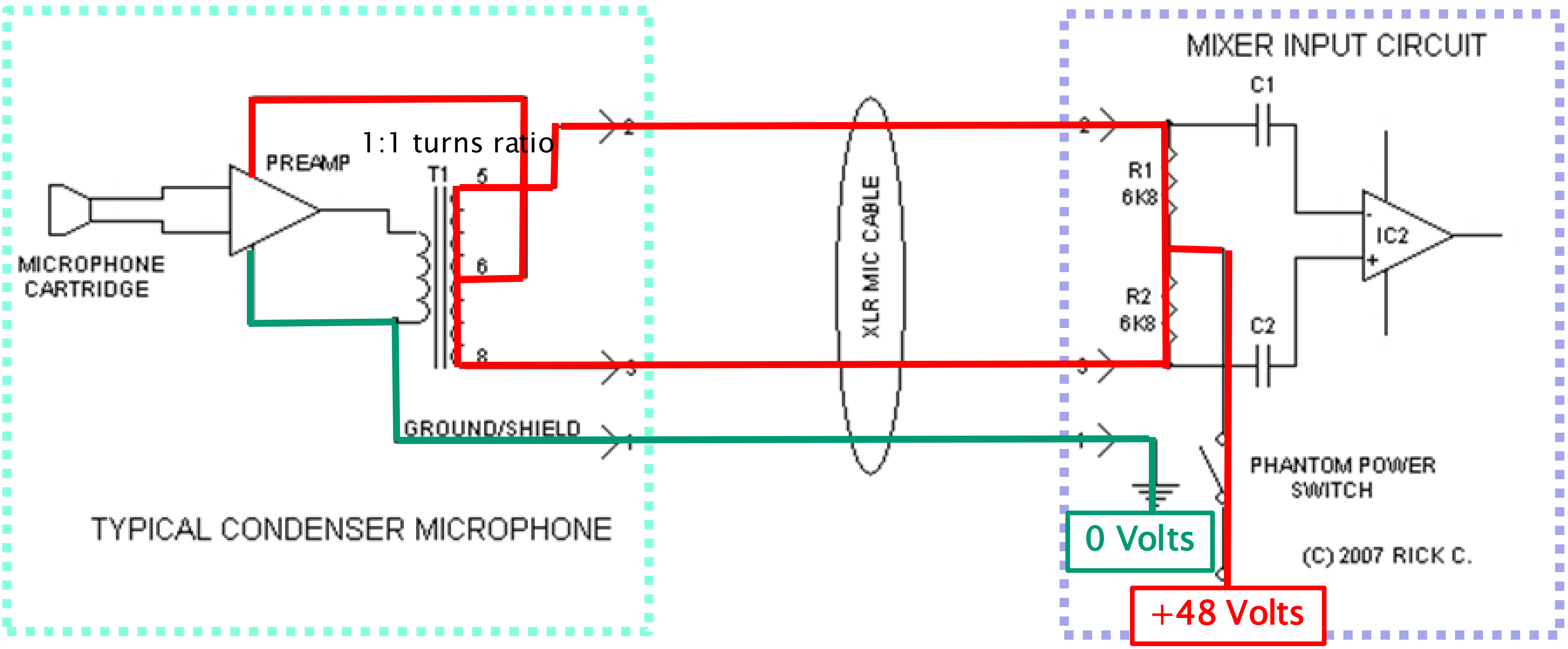
Phantom Power

Electromagnetic Induction Requires a Changing Magnetic Field



Phantom Power – DC Voltages to Feed the Microphone

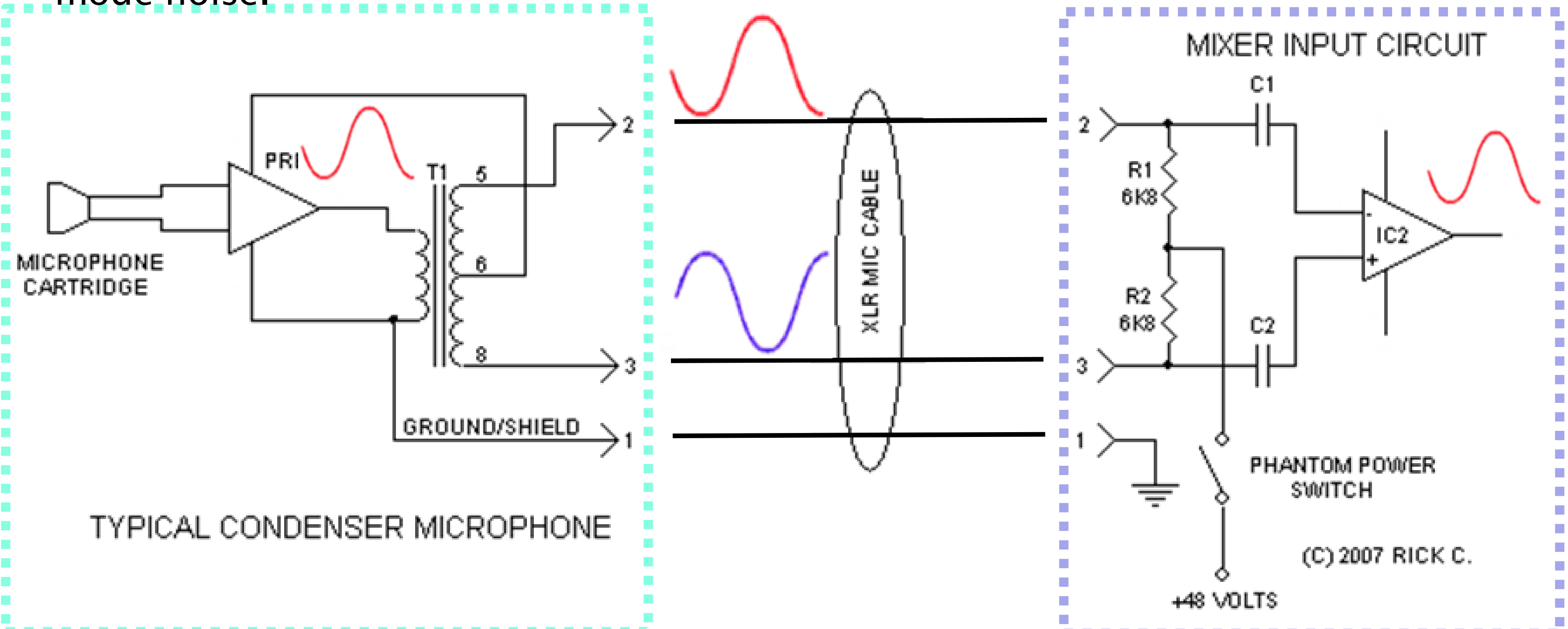
- Power is often required to power up condenser microphones
- DC voltage is applied to the balanced cable without being transferred to the microphone
- The audio signals on the cable are superimposed over the (stable) phantom voltage
- Typical voltage = 48V, resistors typically 6.8kΩ, within 0.4% of each other.



The resistors are matched to ensure the voltage is applied equally to both lines, which prevents any DC offset in the audio signal (since the audio signal is differential).

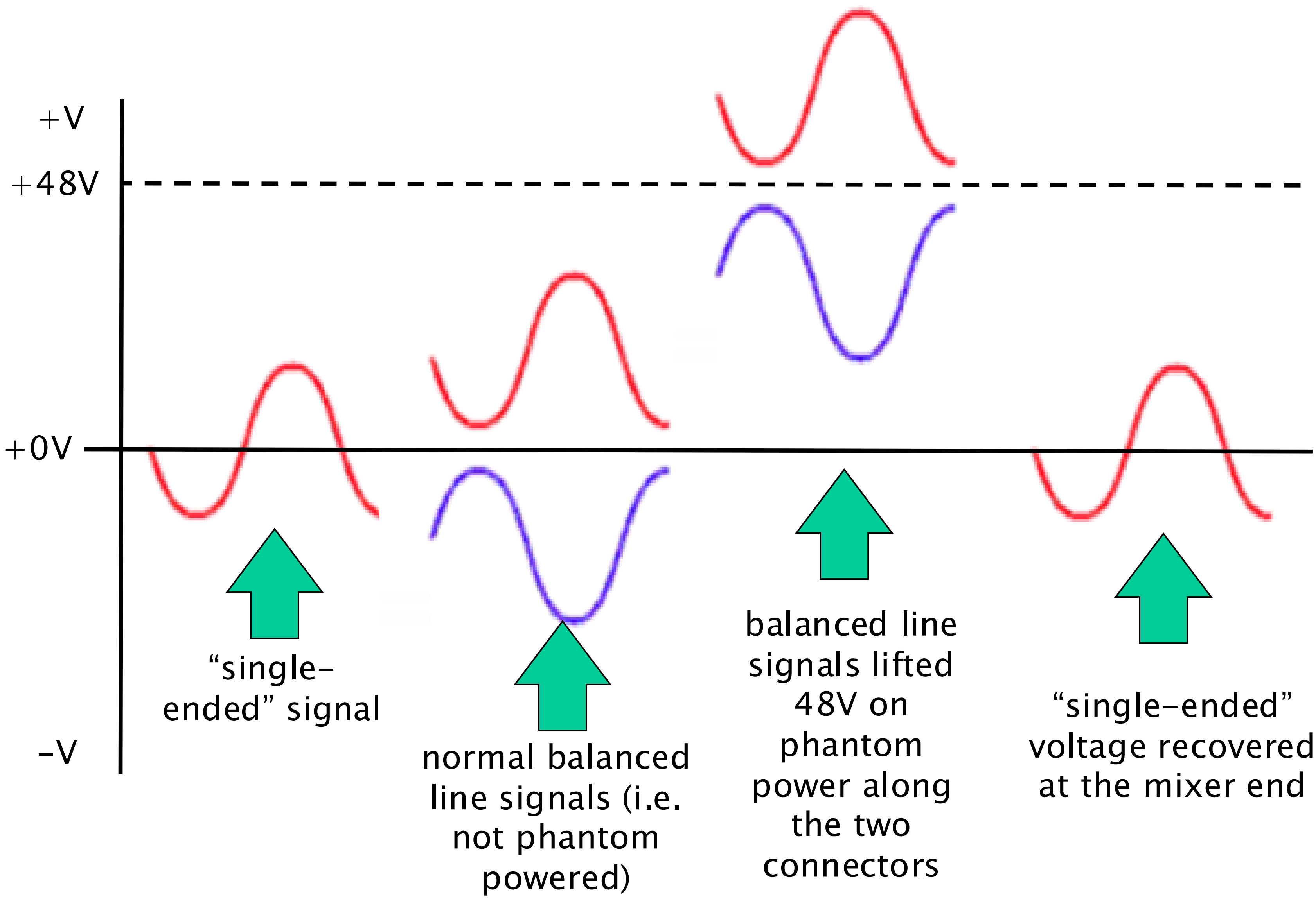
Phantom Power – The Signal Path

- **Microphone Cartridge**: Sound waves are converted into a very small AC electrical signal.
- **Preamp**: This internal amplifier boosts the weak signal from the cartridge to a usable level.
- **T1 (Transformer):** Converting the unbalanced signal from the preamp into a balanced signal on pins 2 and 3 of the XLR connector.
- At the mixer input, coupling capacitors block the DC phantom power.
- Differential amplifier (IC2) amplifies the difference between the in-phase and out-of-phase audio signals, resulting in a stronger output and rejection of common-mode noise.



http://neatcircuits.com/audio/phantom.htm

Phantom Power – Signal Voltage Levels



SDI (Serial Digital Interface), *et al*
… and going forwards (to IP (Internet Protocol))

~ 1990 – Broadcast goes digital

- SDI and other standards (AES (Audio Engineering Society) for audio, etc) for digital media in broadcast

~ 2010 onwards – Broadcast goes IP

- *(why?)*
- The advantages of IP include flexibility, scalability, cost-effectiveness, and the ability to converge video, audio, and data onto a single network.
- but SDI still in development and provision
- ~ 2017 development of standards for support of IP
  - SMPTE ST2022 for SDI into IP: A suite of standards for encapsulating SDI signals for transport over IP networks
  - SMPTE ST2110 for full IP: A comprehensive set of standards defining the transport of professional media over IP networks
  - SMPTE2059 (IETF PTP) for synchronisation: Precision Time Protocol (PTP) is crucial for synchronising media streams in IP-based broadcast systems.
  - AES 10, 67 for audio in IP: Standards for transporting digital audio over IP networks.
  - … more on this later in the course

.. meanwhile, here is SDI …

Serial Digital Interface

SDI standards, developed primarily by SMPTE (Society of Motion Picture and Television Engineers), define how digital video (and often embedded audio) is transmitted over coaxial cables, typically with BNC connectors.

Family of interfaces
- SMPTE ITU-R BT.656, SMPTE 259M – broadcast-grade video
- HD-SDI – SMPTE 292M – nominal data rate = 1.485 Gb/s

Recent interfaces for
- HD, UHD, higher frame rates, 3D, increased colour depth

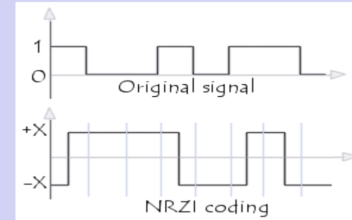
Dual link HD-SDI
- pair of SMPTE 292M links (SMPTE 372M) = 2.970 Gbs
    - digital cinema, 2K (2048x1080)
    - HDTV 1080P

Uncompressed, unencrypted digital video
- may include embedded audio + time code
- also for packetized data

SDI and HD-SDI usually only in professional video
- due to licensing restrictions of unencrypted digital interfaces
- alternative – consumer cameras – HDMI

Serial Digital Interface

NRZI
bit = 1, the signal changes state after the clock tick.
bit = 0, the signal does not change state.



http://ccm.net/contents/703-data-transmission-digital-data-transmission

Electrical / Physical

- coaxial cables, BNC connectors
- nominal impedance of 75Ω for the coaxial cable and connectors
- signal amplitude at source = 800 mV (±10%) p-p
- 270 Mbps SDI over 300 m; but shorter lengths preferred
- length of transmission typically <300 m
- fibre variants – eg 297M for long-distance transmission
- HD bitrates – typically 100 meters

- Uncompressed digital component signals
- NRZI format
- data scrambled to reduce long strings of 0's and 1's
- SDI signal is designed to be self-synchronizing, meaning the receiver can extract the clock signal directly from the data stream, eliminating the need for a separate clock signal. This is achieved through the NRZI encoding and data scrambling.

SMPTE 2022

- digital video over IP network, includes MPEG-2 (systems) and SDI

Serial Digital Interface

Data Format

- SD: each color component (Y, Cb, Cr) is represented by 10-bit samples.
- HD: 20b: 2 x 10-bit datastreams (10b Y then 10b C= luminance (Y) samples are transmitted first, followed by the chrominance (Cb and Cr) samples.)
- chrominance blue-difference (Cb) and chrominance red-difference (Cr) carry the colour information of the video signal.
- luminance channel (Y): Carries the brightness information of the video signal.

Datastream

- color encoding 4:2:2 YCbCr, a standard for digital video.
- SD: … Cb Y Cr Y' Cb Y Cr Y’ …
- HD: … Luma – Y Y' Y Y' Y Y' Y Y’ … Chroma – Cb Cr Cb Cr Cb Cr Cb Cr

- luminance channel (Y)
  - Occupies full bandwidth (13.5 MHz @ 270 Mbit/s SD; ~75 MHz HD)
- chrominance channels (Cb and Cr)
  - subsampled horizontally, encoded at half bandwidth (SD: 6.75 MHz or HD: 37.5 MHz).

Video payload

- 10b word: In a 10-bit SDI signal, the valid range for the video data values is typically from 4 to 1,019. (2^10=1024)
- 0-3 and 1,020-1,023 reserved – Synchronization, other control data

Serial Digital Interface

Line counter and Cyclic Redundancy Check (CRC)

- add additional check words: SDI includes extra data to help detect errors that might occur during transmission.
    - HD: CRC field = CRC of the preceding line

      .. computed independently for the Y and C streams

Line and sample numbering

- Each sample within a datastream is assigned a unique line and sample number
- Lines numbered sequentially
    - starting from 1, up to the number of lines per frame of the indicated format

      … typically 525, 625, 750 (1125 for Sony HDVS)

Link numbering – for multi-link interfaces

- When multiple SDI cables are used together to achieve higher bandwidth (as in Dual-Link HD-SDI), the first cable is designated as the primary link and assigned the number 1.
- subsequent links are assigned increasing link numbers

AES 3

AES/EBU (Audio Engineering Society/European Broadcasting Union)

- digital audio signals
- two channels Pulse Code Modulation (PCM) audio
- balanced lines, unbalanced lines, and optical fiber
- 1985, revised in 1992 and 2003

IEC 60958 Type I—Balanced, XLR

- 3-conductor, 110-ohm twisted pair
- professional installations

IEC 60958 Type II—Unbalanced, RCA

- unbalanced, 2-conductor, 75-ohm coax, RCA connectors (phono)
- consumer, coaxial S/PDIF

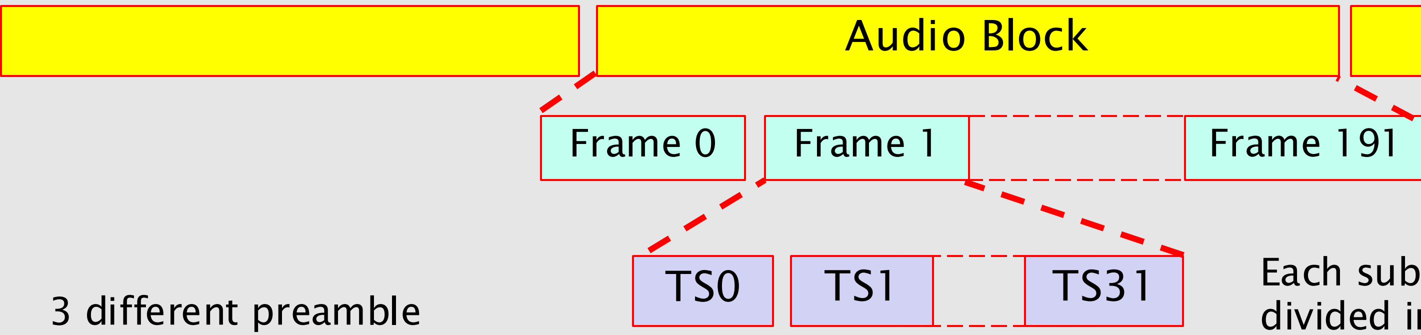
IEC 60958 Type III Optical—Fiber, F05/TOSLINK

- optical fiber, typ plastic
- Toshiba brand name, TOSLINK
- consumer, optical S/PDIF

AES-3id

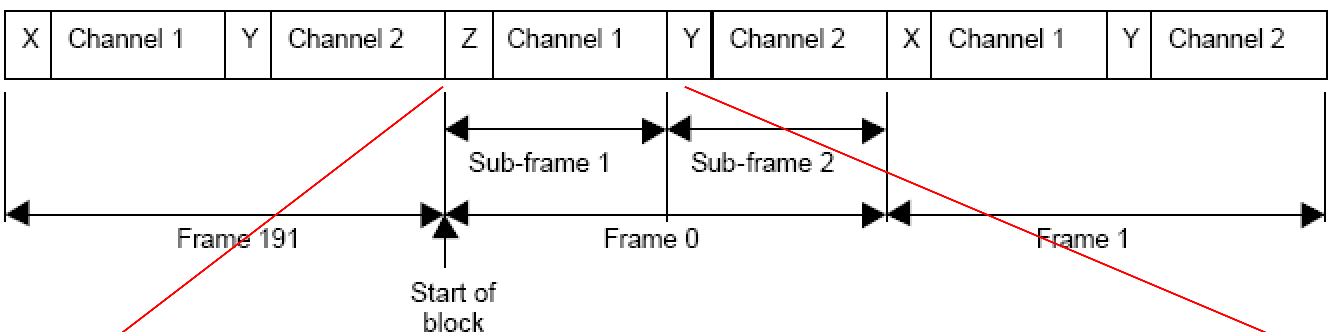
- 75-ohm BNC electrical variant of AES3
- analogue or digital video
- common in the broadcast industry

AES 3 Framing



Each subframe is further divided into 32 time slots (TS)

3 different preamble patterns (X, Y, and Z)



Each frame contains two subframes, one for each audio channel (Channel 1 and Channel 2).

subframes:

32-bit subframe

Preambles help the receiver identify the start of a subframe and the frame number within the audio block.

when sample is 20bit coded …